

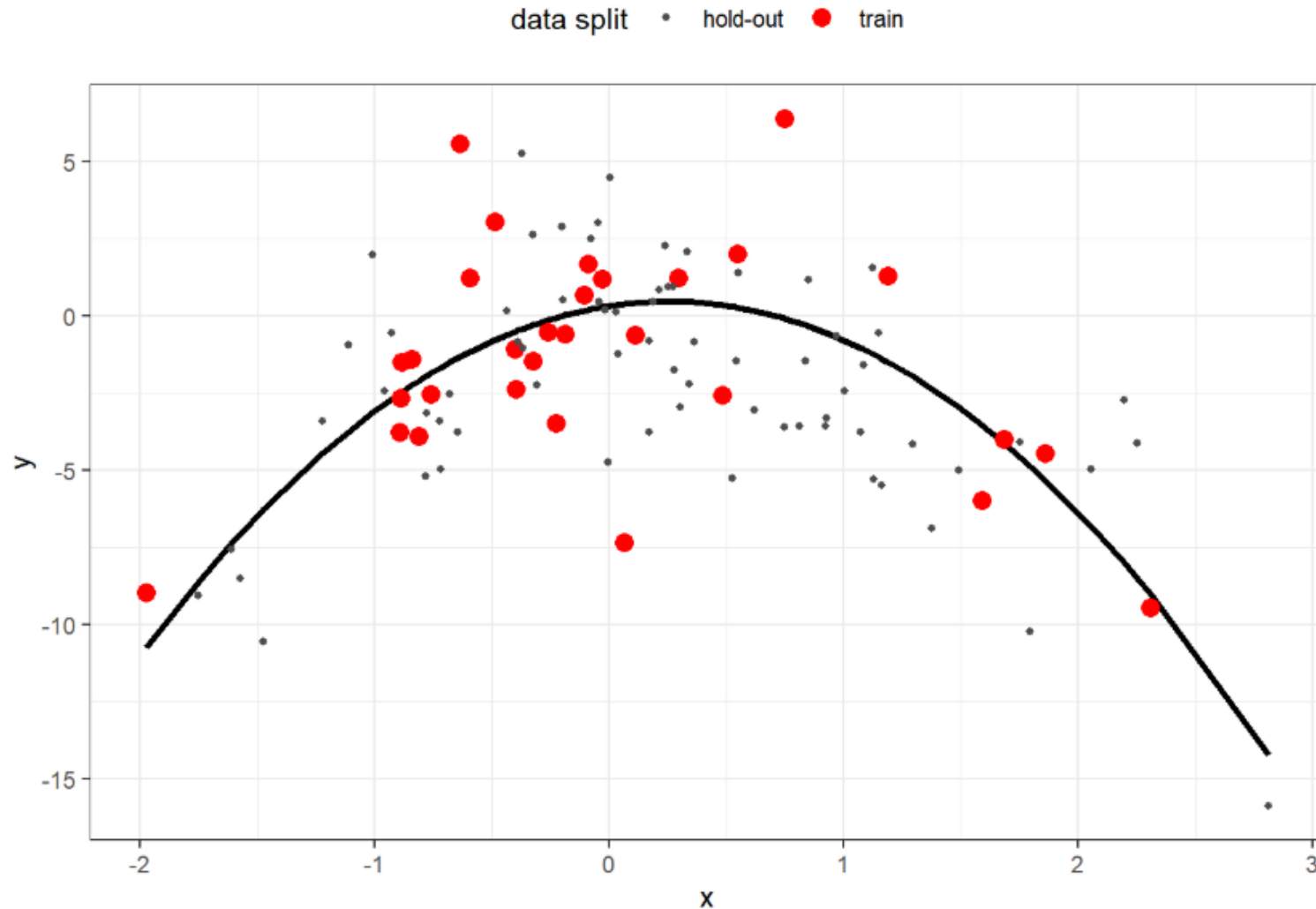
We know how to generate many different kinds of features from inputs!

- Continuous inputs
 - Linear features (MAIN EFFECTS)
 - Basis functions (polynomials, sin/cos, splines, etc...)
 - Interactions – products between continuous features
- Categorical inputs
 - DUMMY variables or One Hot Encoding
- Mixed inputs
 - ADDITIVE or INTERACTIONS between continuous and categorical inputs

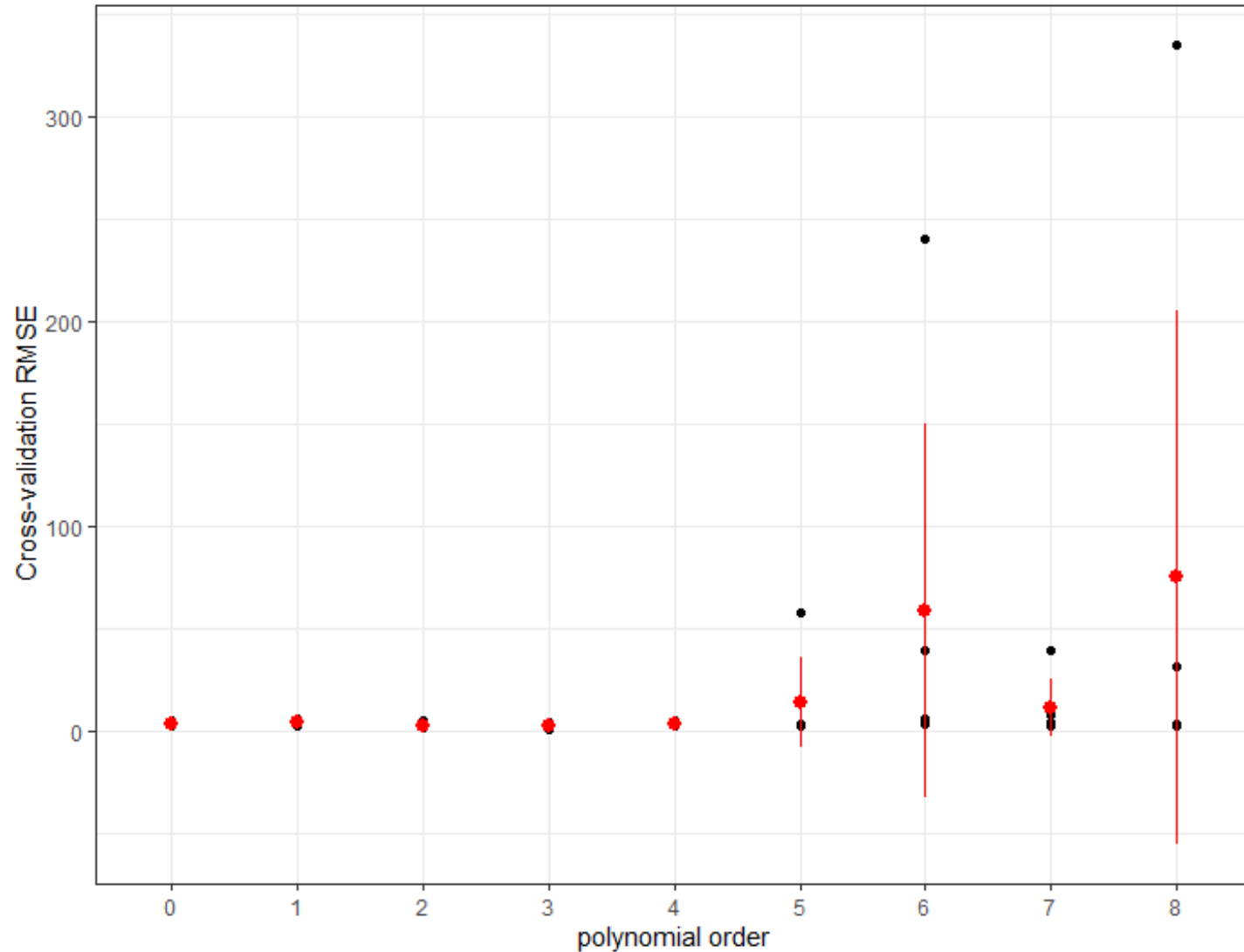
We have seen two different ways to assess if a model is overfitting to a training set

- We have used **cross-validation** to get an idea about the generalization error on hold-out sets via **resampling**.
- We have used the Laplace Approximation to estimate the **log-Evidence** which penalizes the performances essentially based on the number of parameters in the model.
 - We discussed how we can approximate the **log-Evidence with the BIC**.

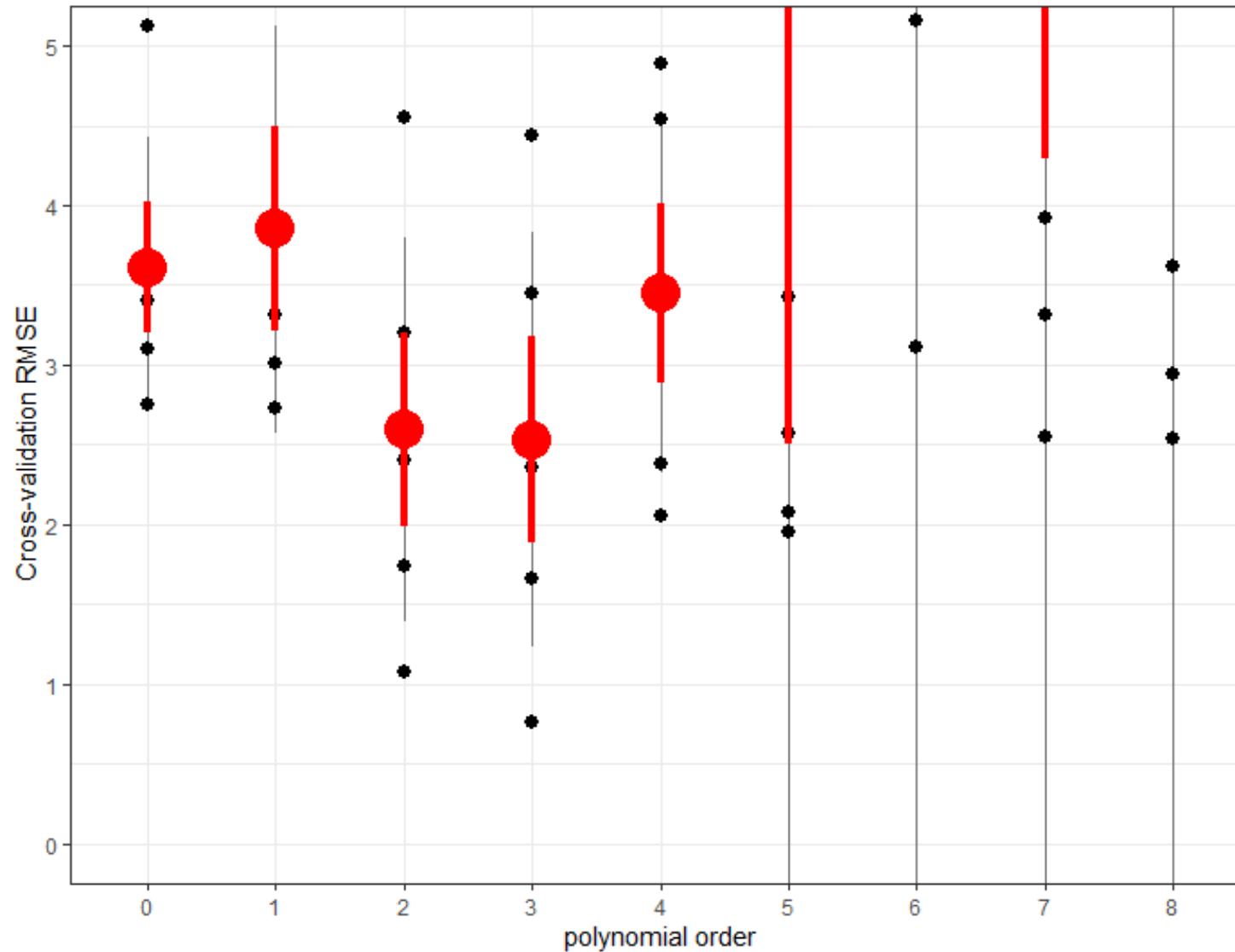
Let's return to the noisy quadratic demo we used when we introduced cross-validation



5-fold cross-validation RMSE results



Zoom in on the 5-fold cross-validation results

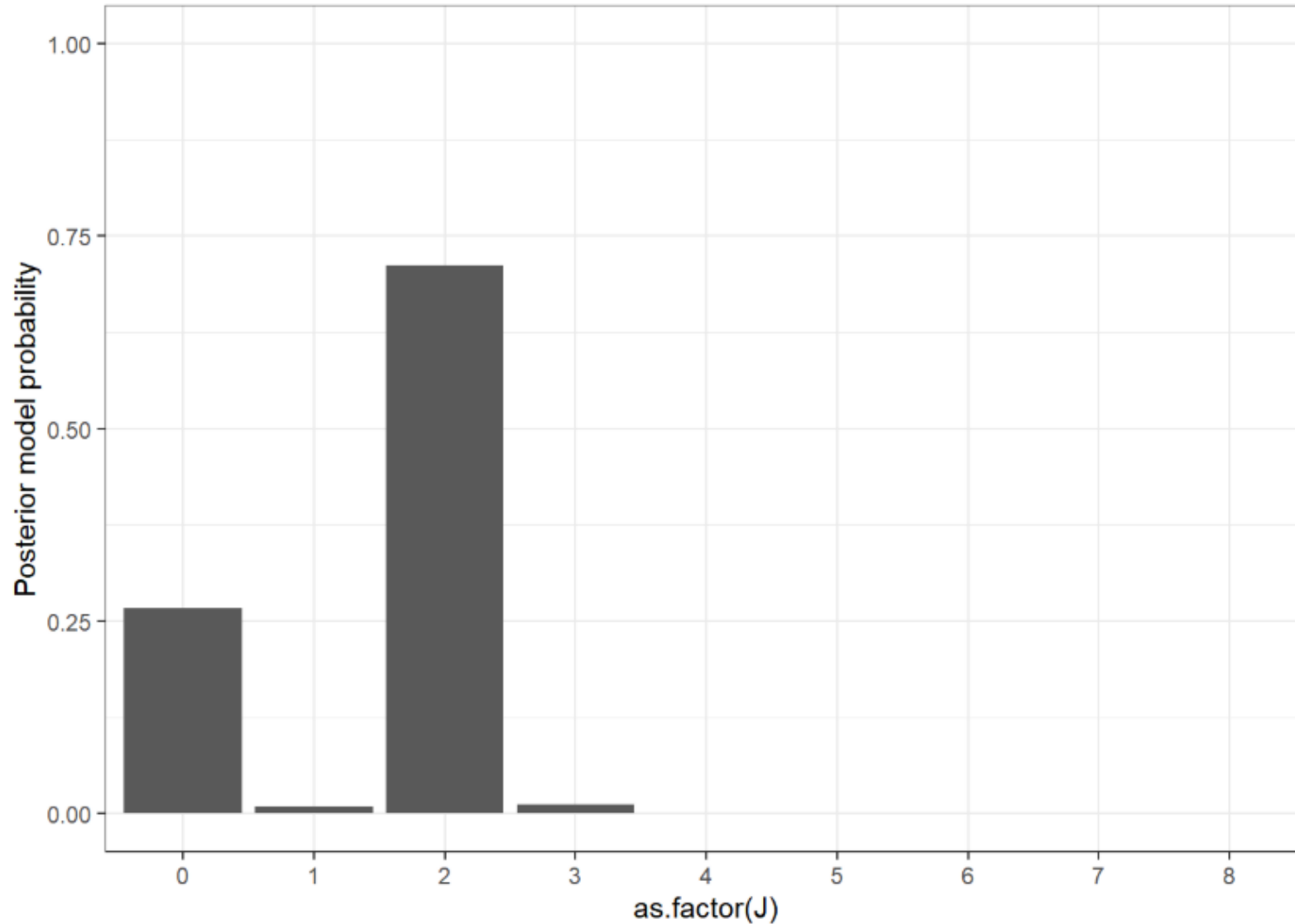


Quadratic and cubic models are effectively indistinguishable from each other!

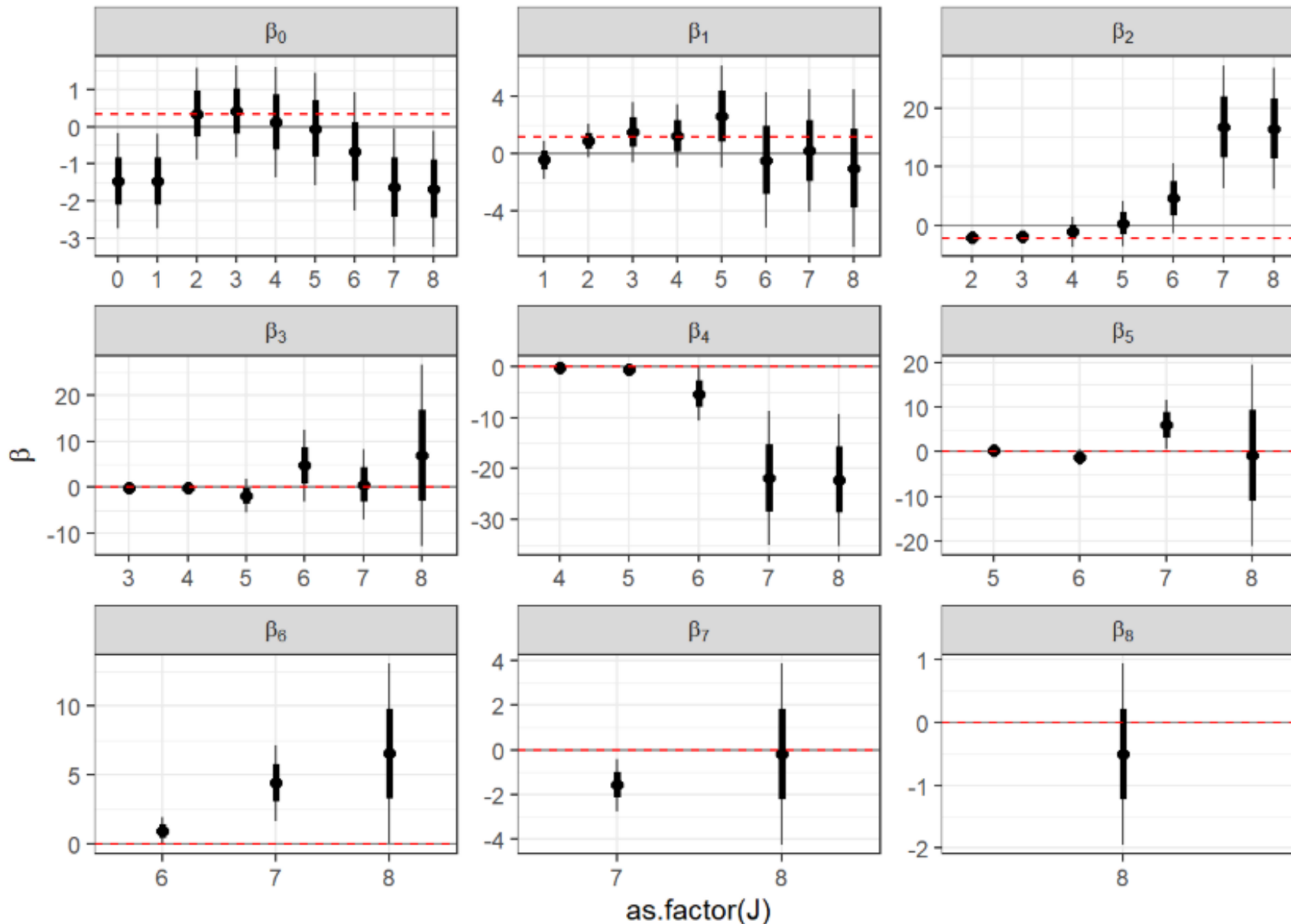
Fit Bayesian linear models to the training set with the Laplace Approximation

- Assume a shared prior standard deviation on the coefficients of $\tau_{\beta} = 25$ and use the prior rate on σ to be $\lambda = 1$.
- **Fit all 9 models and compare the models with the log-evidence.**
- Which model will be considered to be the most generalizable?

Quadratic model is properly identified!



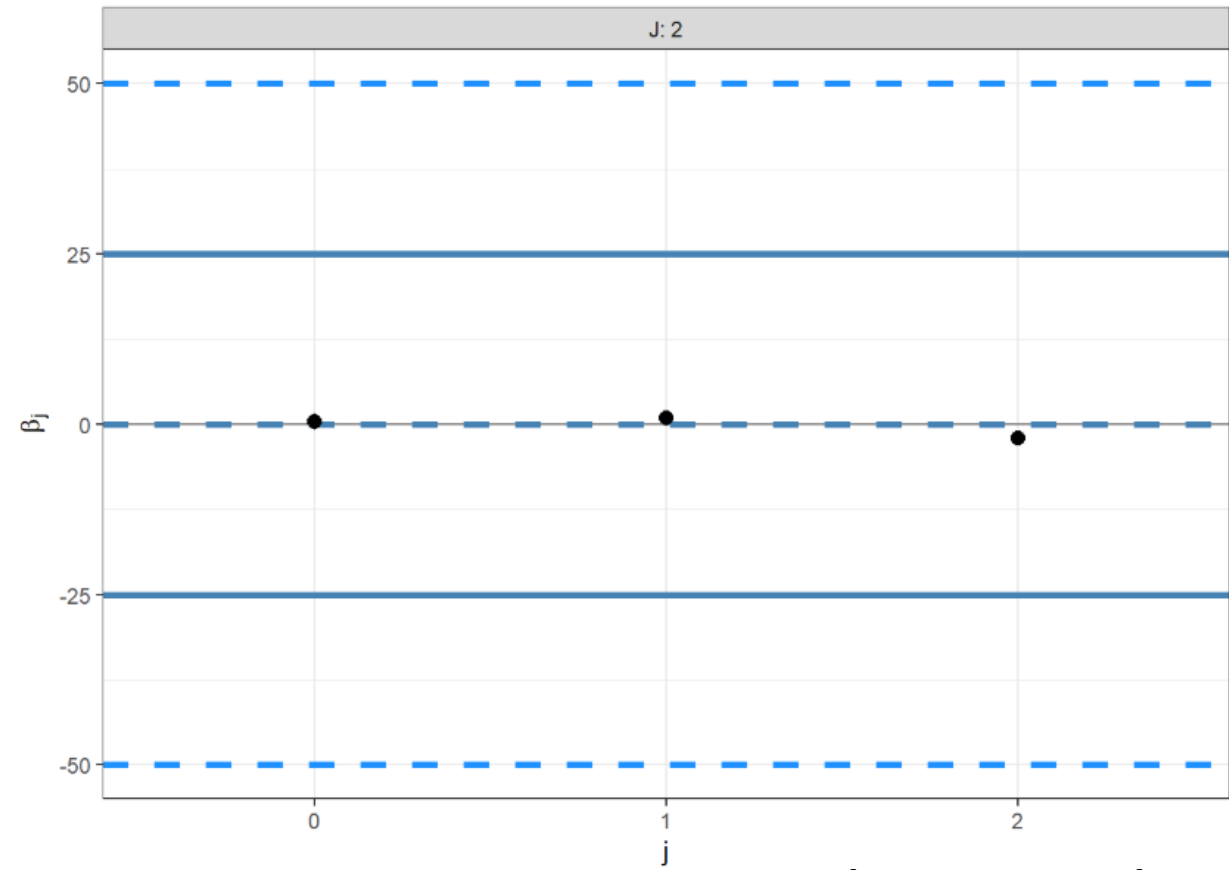
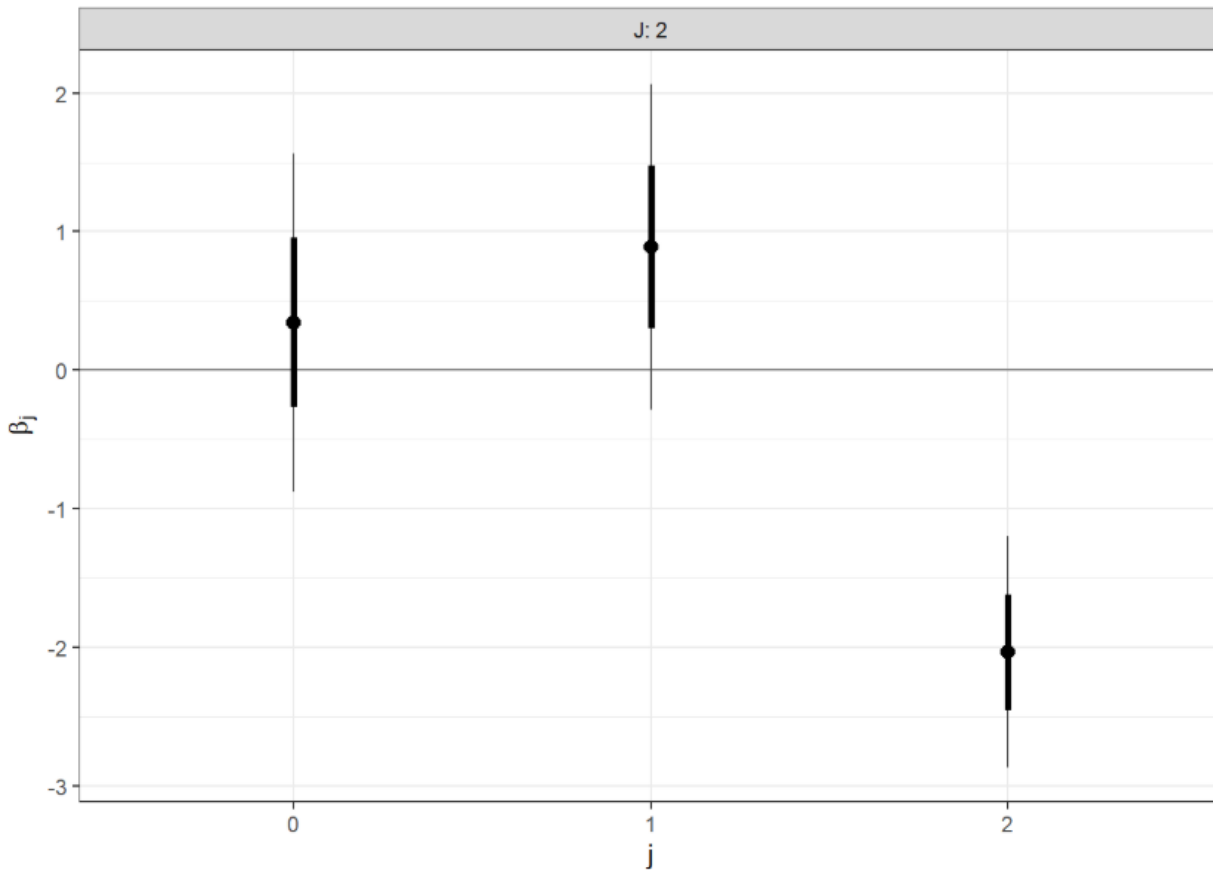
Take a look at the coefficients



Higher order polynomials have large or extreme values relative to the lower polynomials.

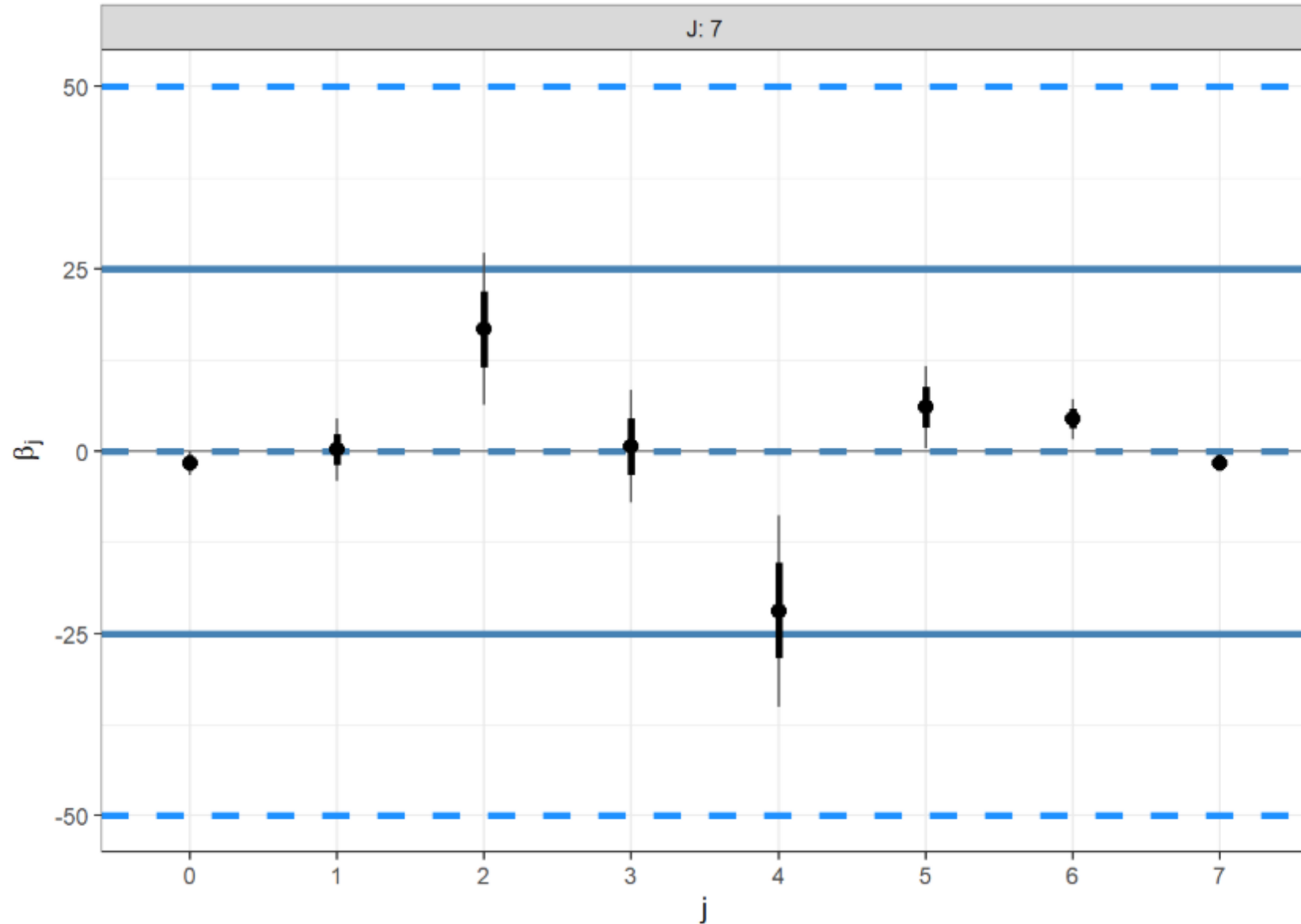
Vertical axes bounds are different across the subplots.

Focus on the quadratic model

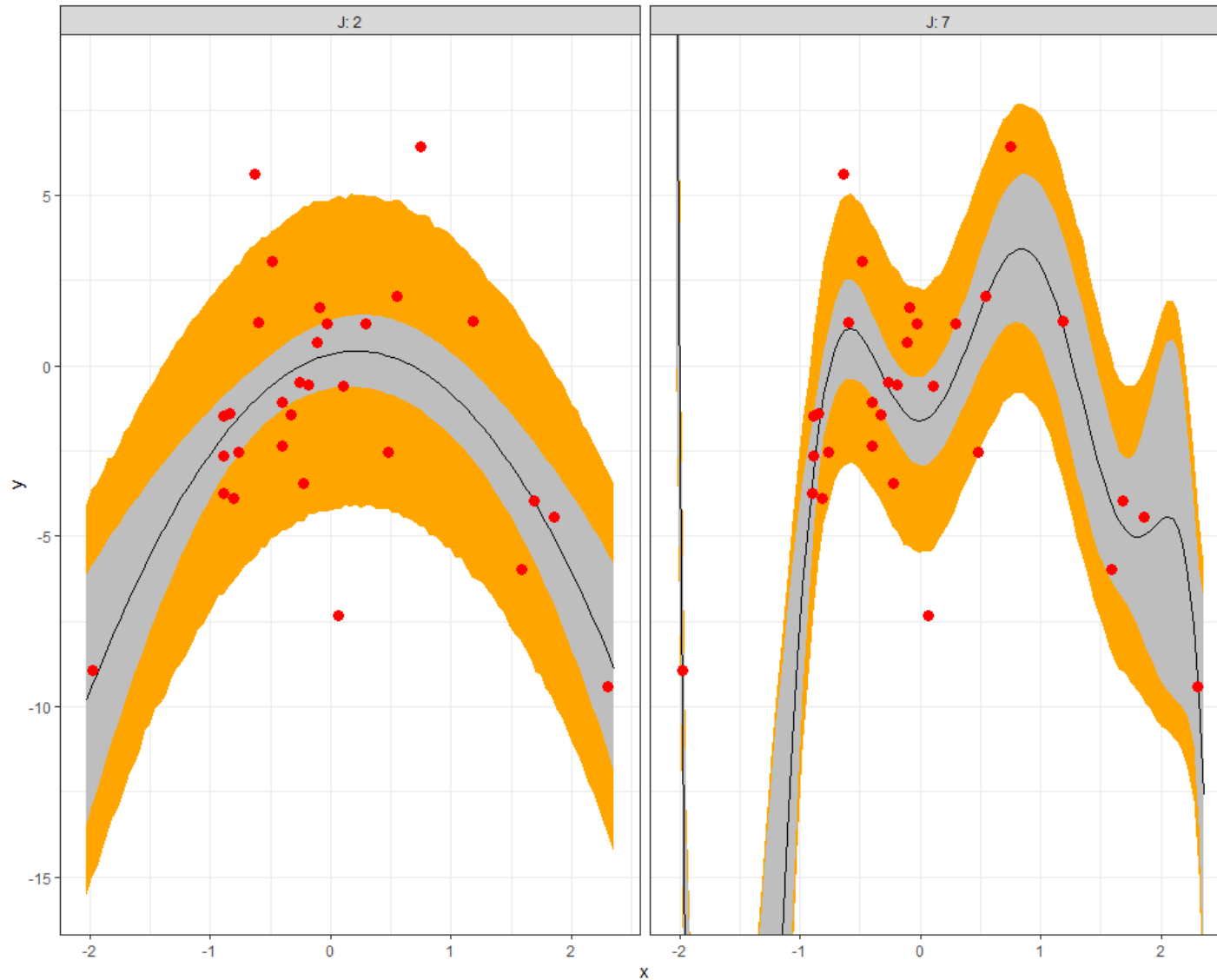


Posterior summaries relative to the prior 95% uncertainty interval

Contrast with the behavior of the 7th degree polynomial



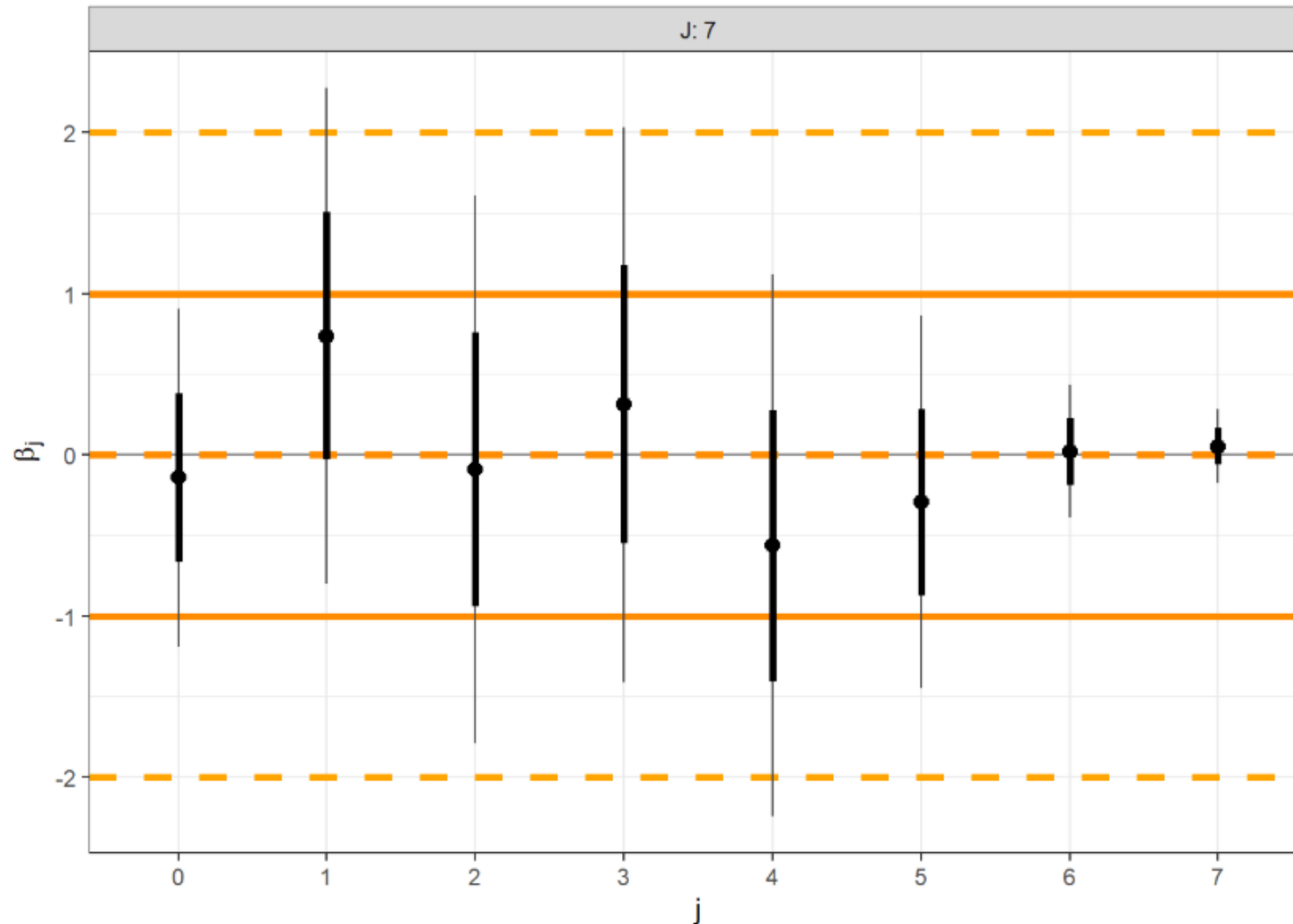
Compare predictions from the quadratic and 7th degree polynomials...we know the more complex model is overfit!



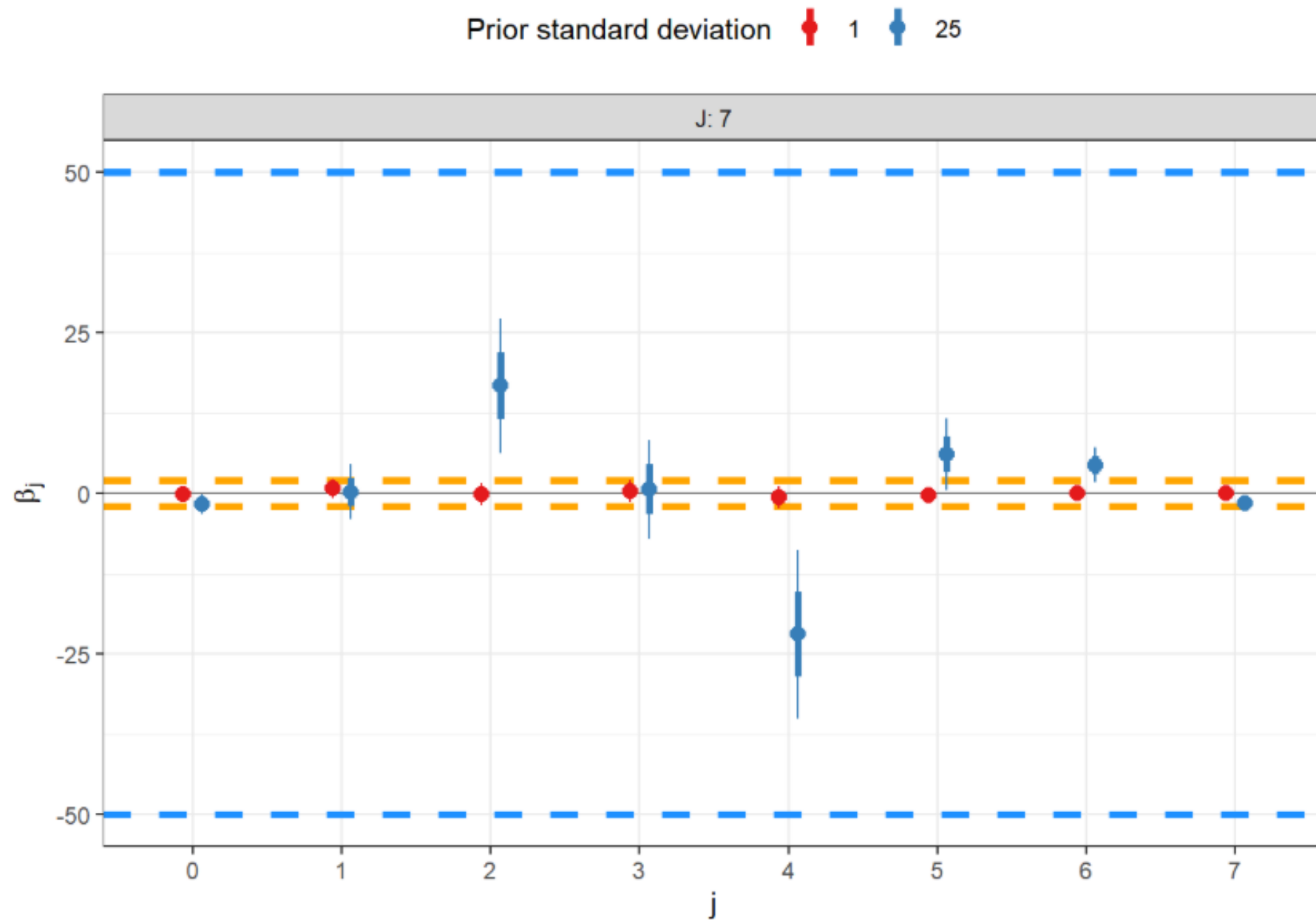
What would happen if we would use an informative or “strong” prior, $\tau_{\beta} = 1$?

- Would we anticipate the posterior to change for the 7th degree polynomial?

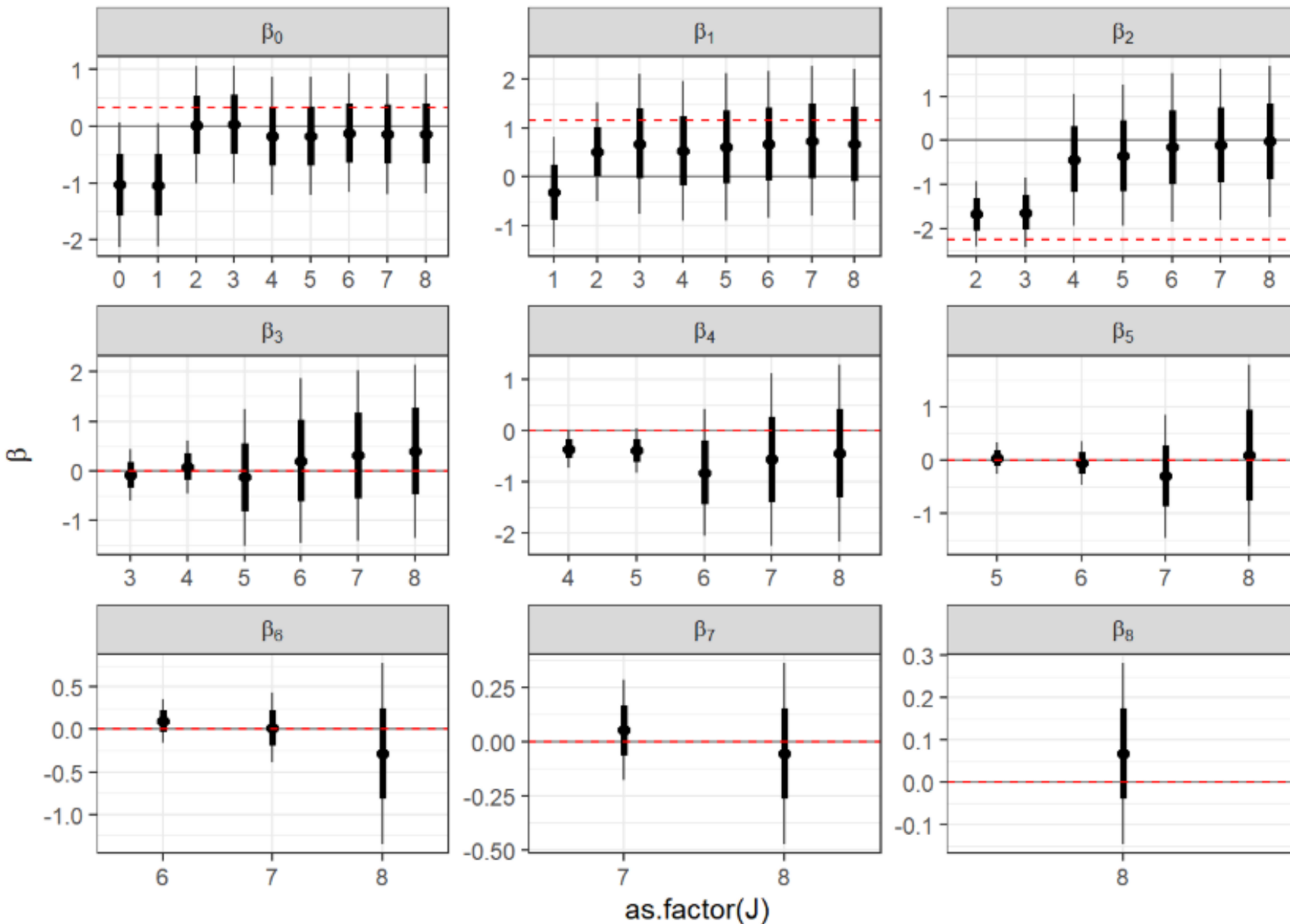
Posterior summaries on the 7th order polynomial coefficients relative to the informative prior 95% uncertainty interval



The informative prior has constrained the parameter values away from the EXTREME values.



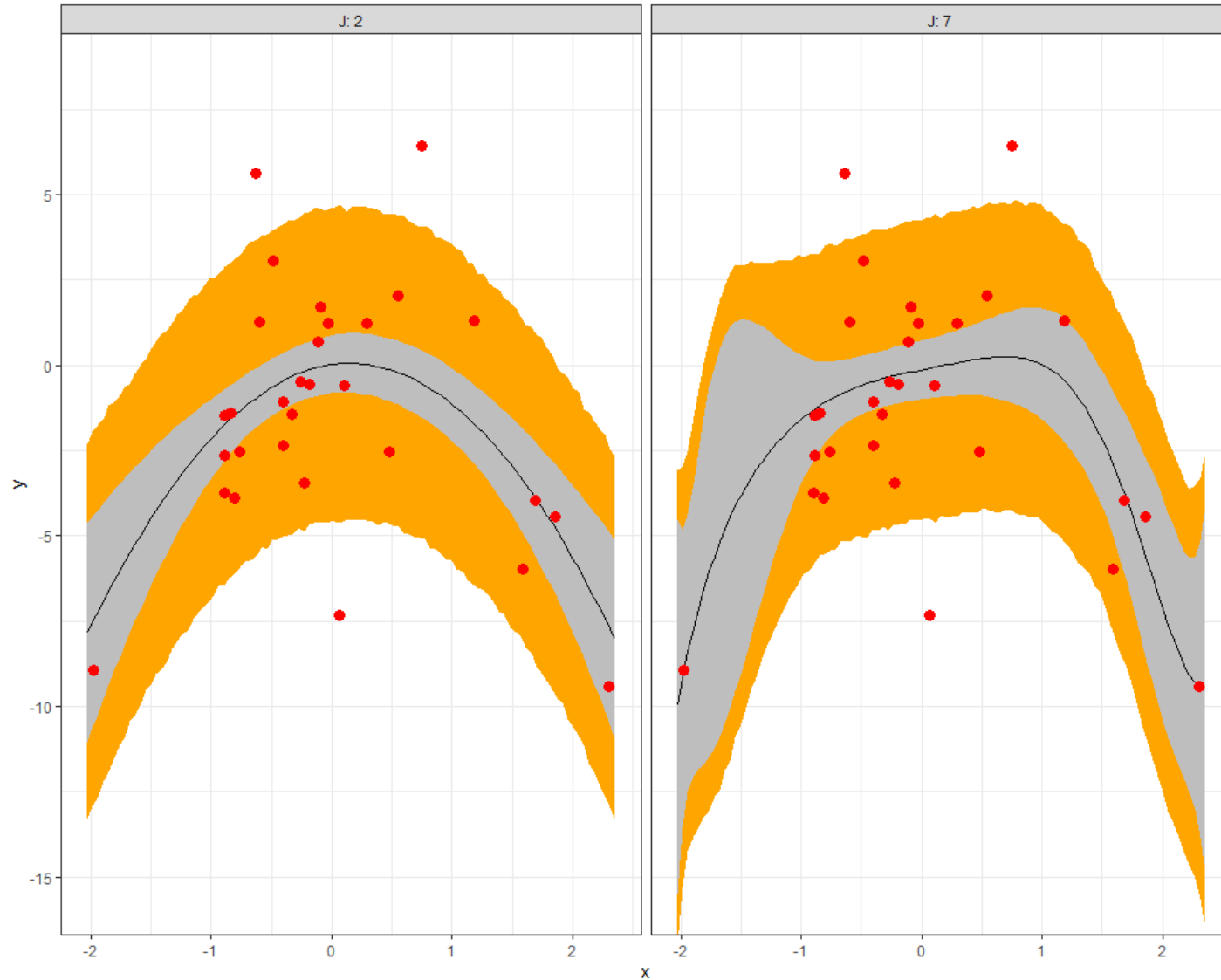
Apply the informative prior $\tau_\beta = 1$ to all polynomials



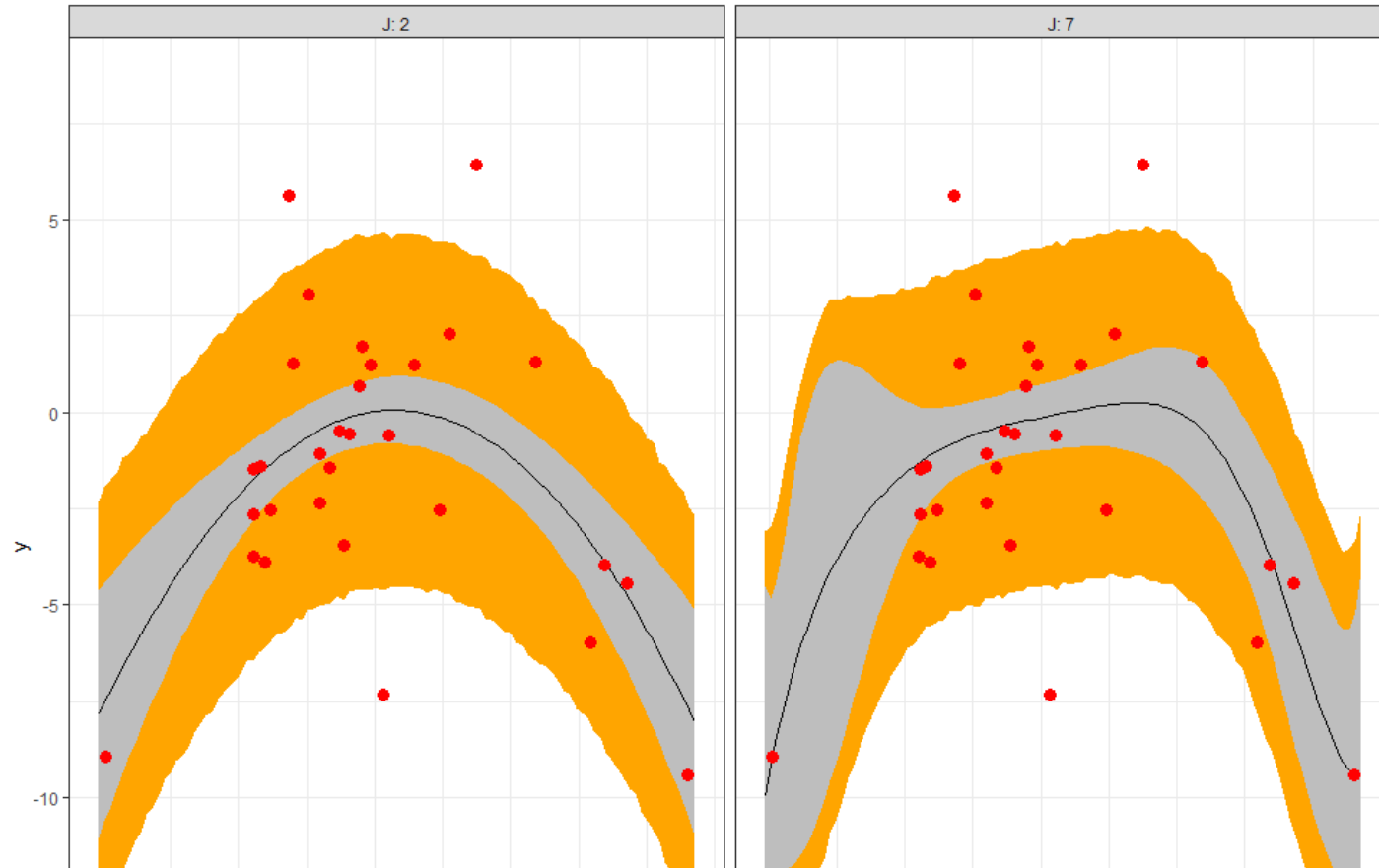
The coefficient posteriors appear to “stabilize” across the higher order polynomials compared to the diffuse prior case.

Vertical axis bounds are different across subplots, but all range between single digits.

Compare a quadratic and 7th degree model based on the informative prior

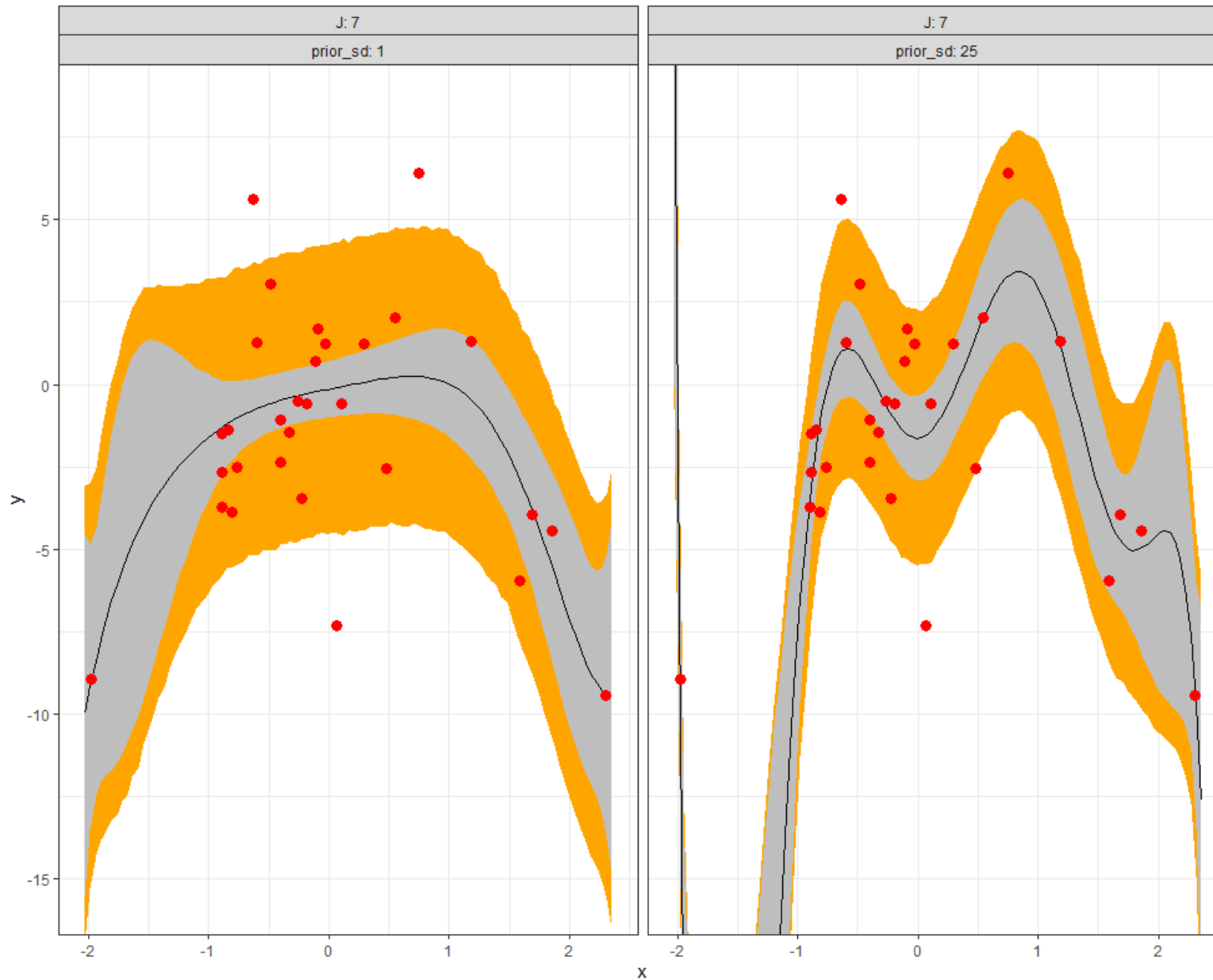


Compare a quadratic and 7th degree model based on the informative prior



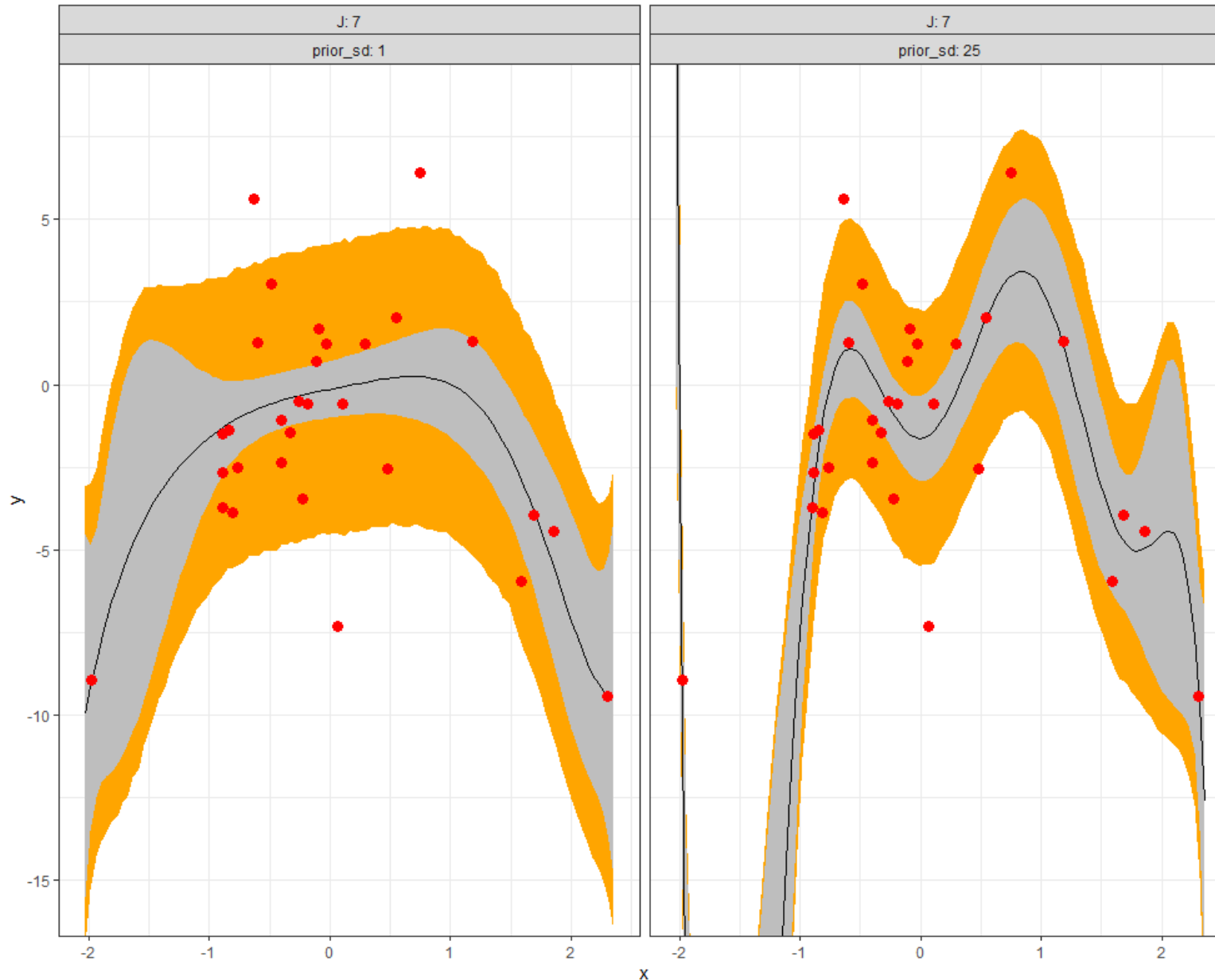
With the informative prior, the 7th degree model “looks” more like the quadratic!!!!

Contrast the 7th degree model between the two priors



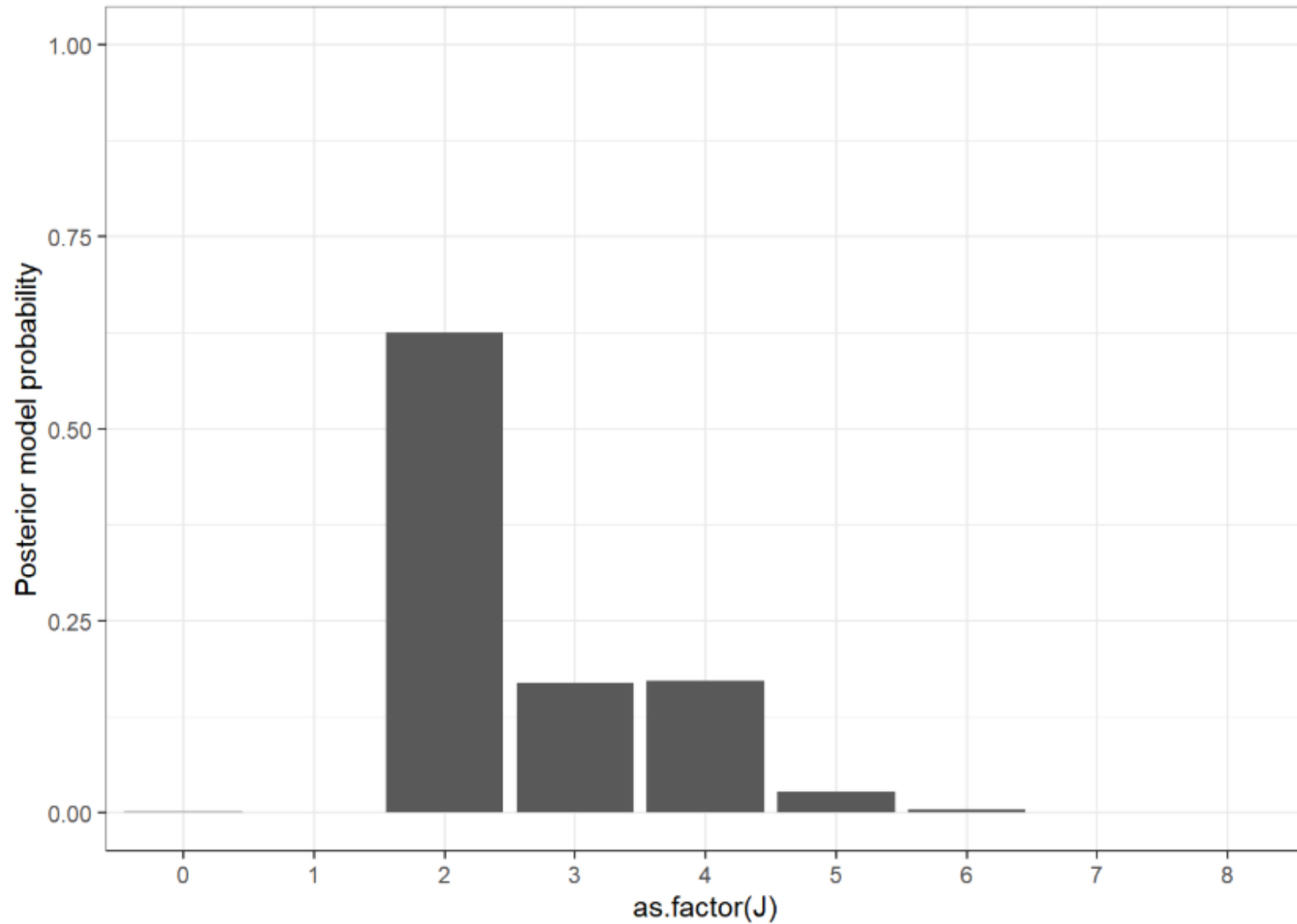
Contrast the 7th degree model between the two priors

Informative prior prevents the 7th degree model from interpolating noise.

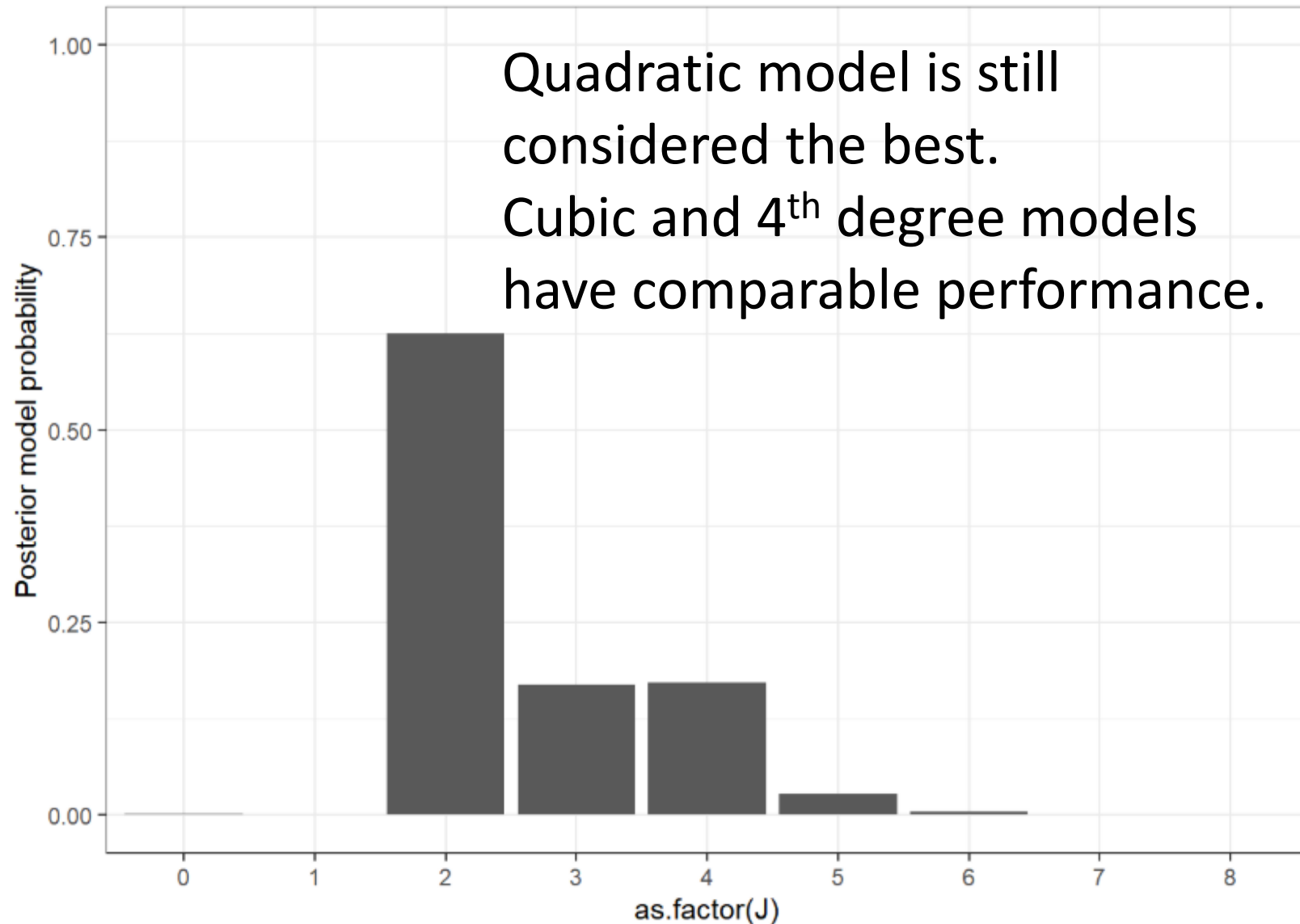


Diffuse or weak prior allows the model the flexibility to try and fit the noisy observations more accurately.

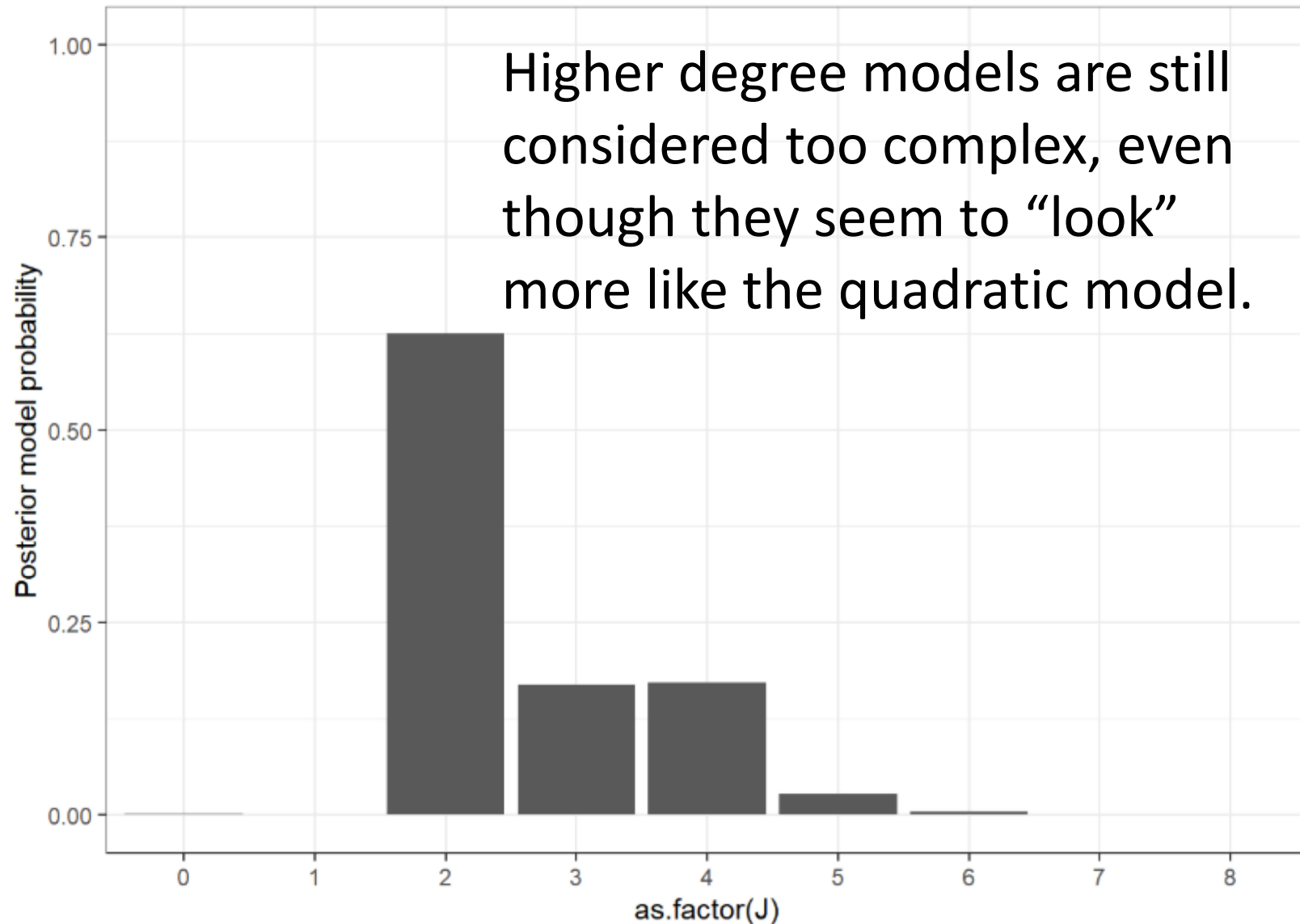
Which model is better under the informative prior?



Which model is better under the informative prior?

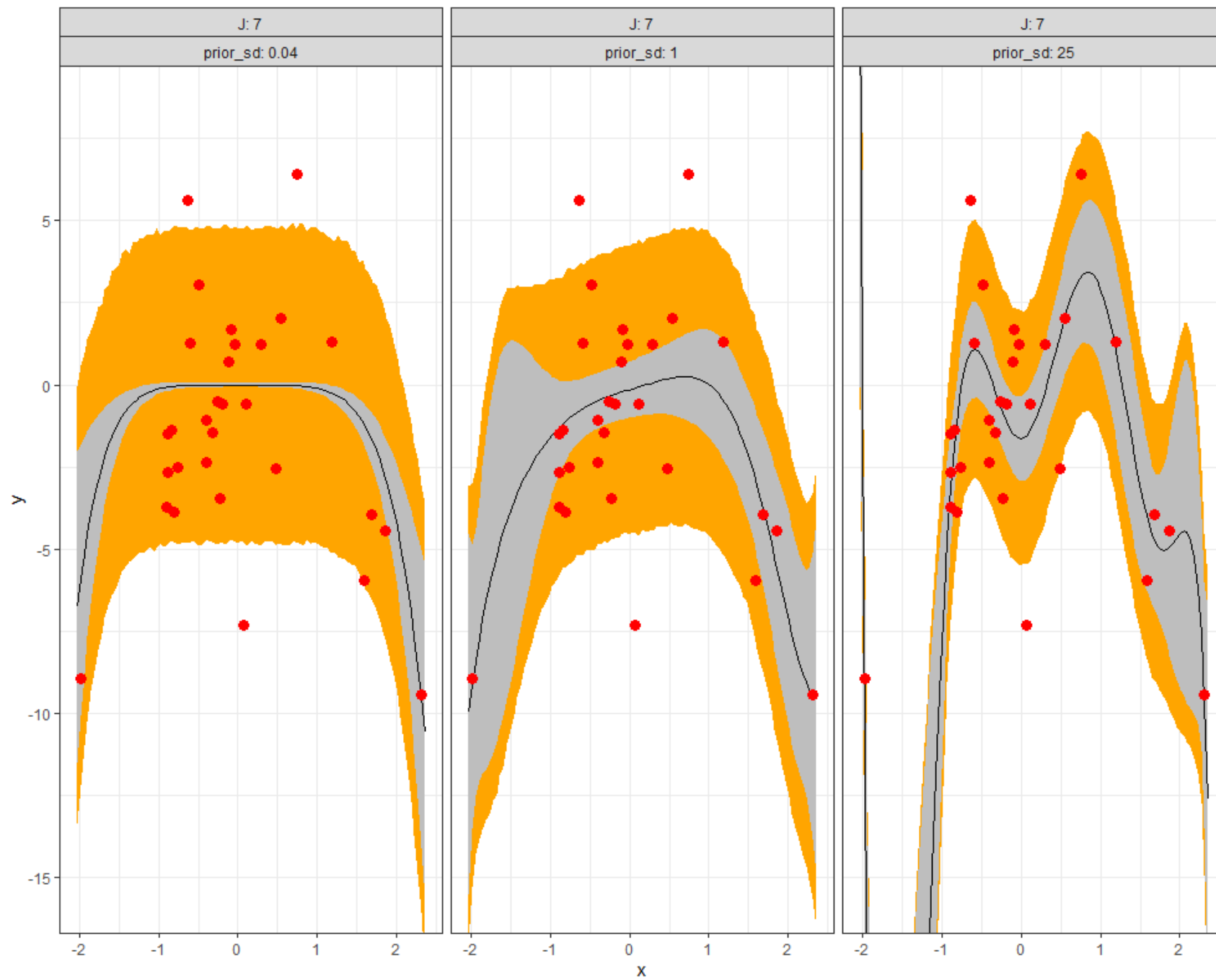


Which model is better under the informative prior?



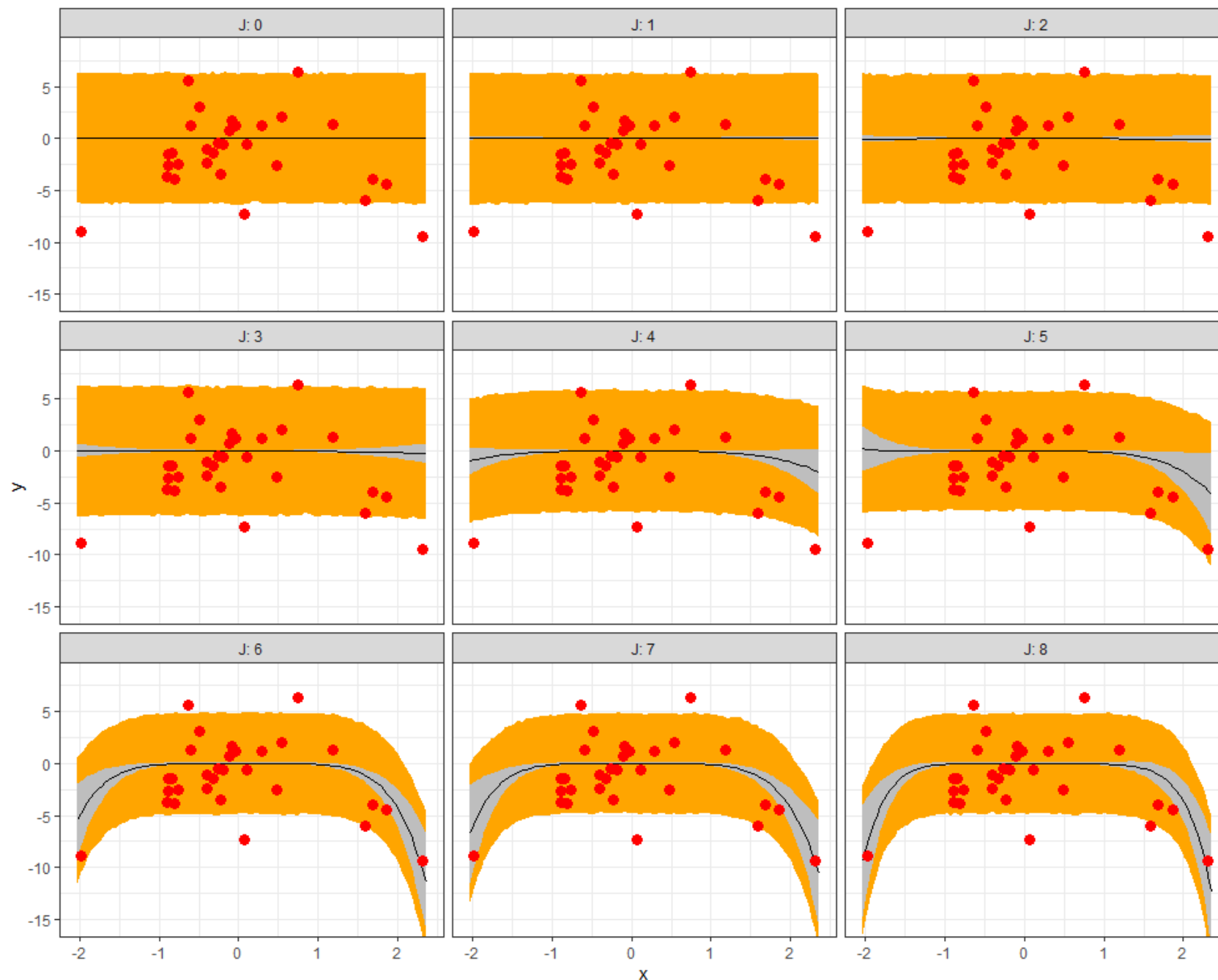
What happens if the prior is VERY strong?

- Set $\tau_\beta = 0.04$...what will happen?

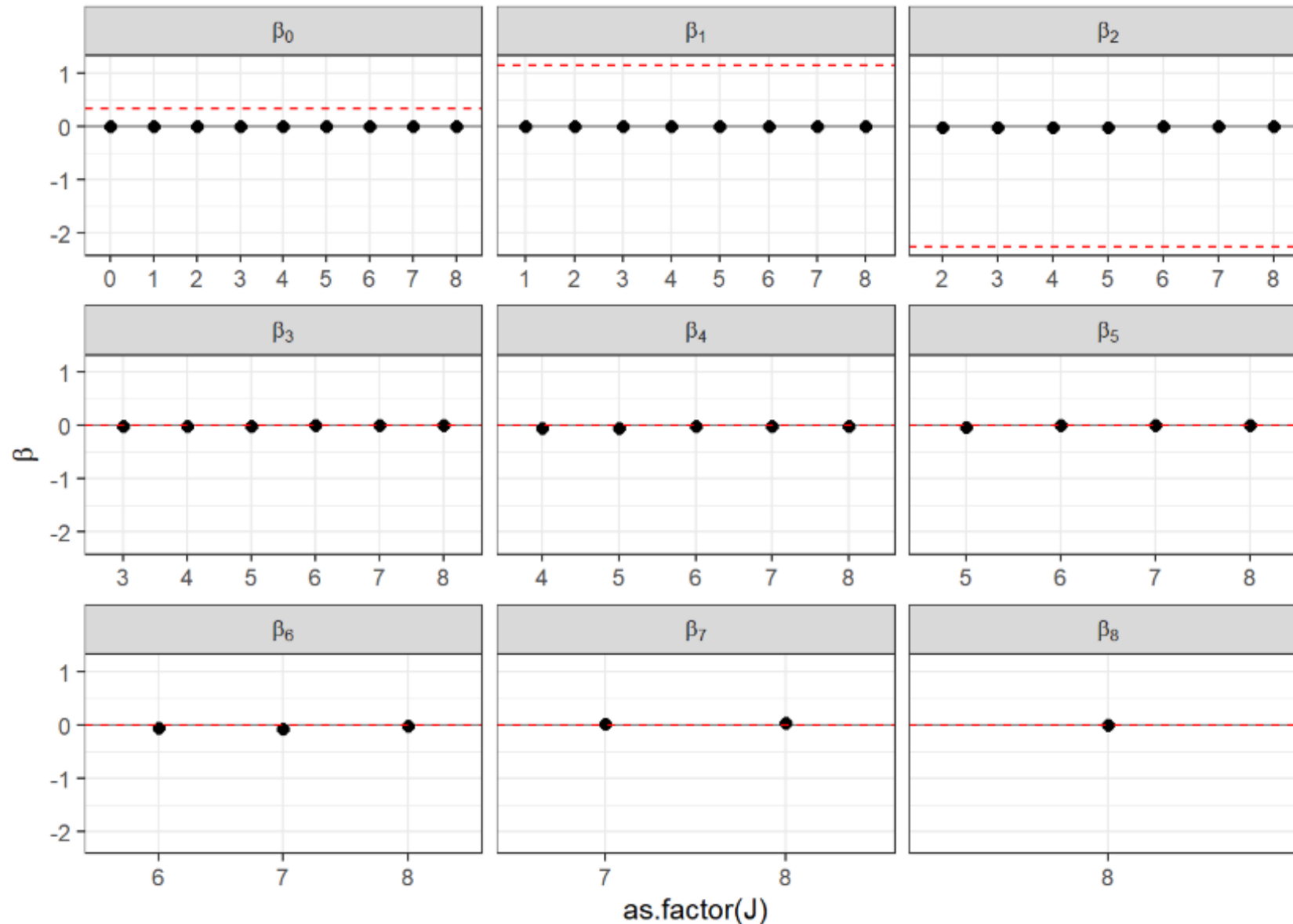


Look at predictions from
ALL polynomials based on
the VERY STRONG prior

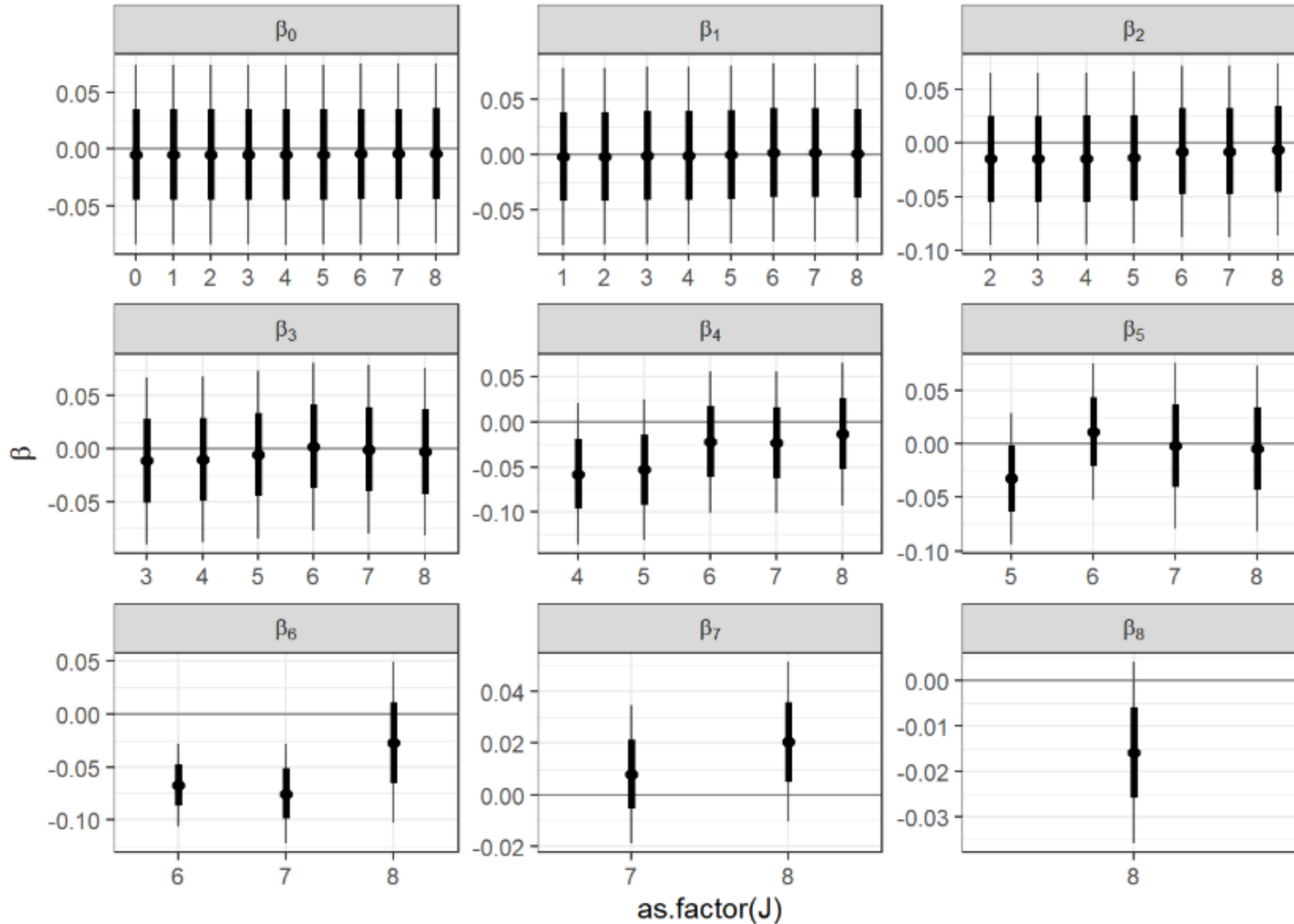
The quadratic model
behaves as a constant
without any trend!!



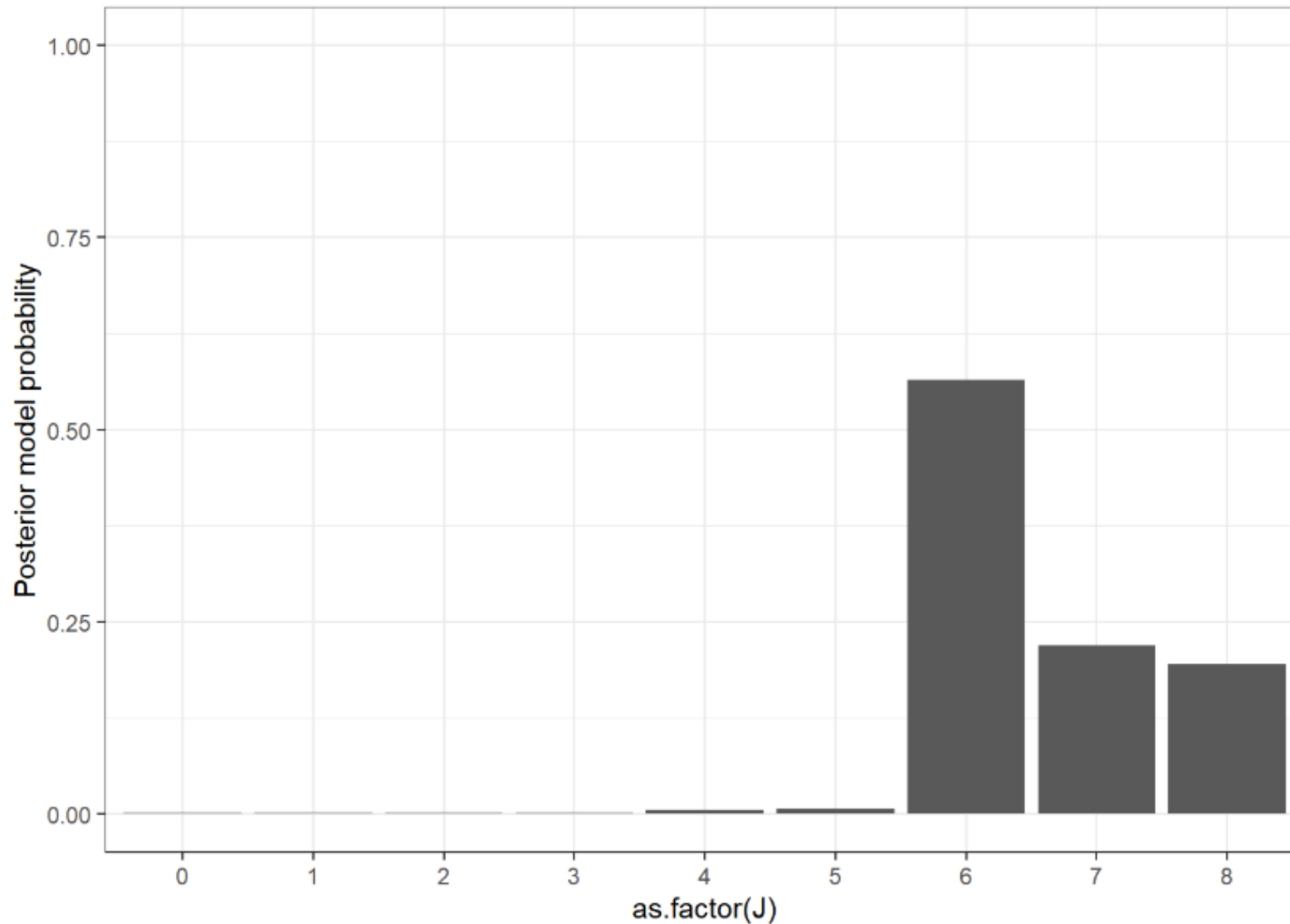
With the very strong prior, we've forced all of the parameters to be very close to zero!



Zoom in to show the parameters are not identically zero



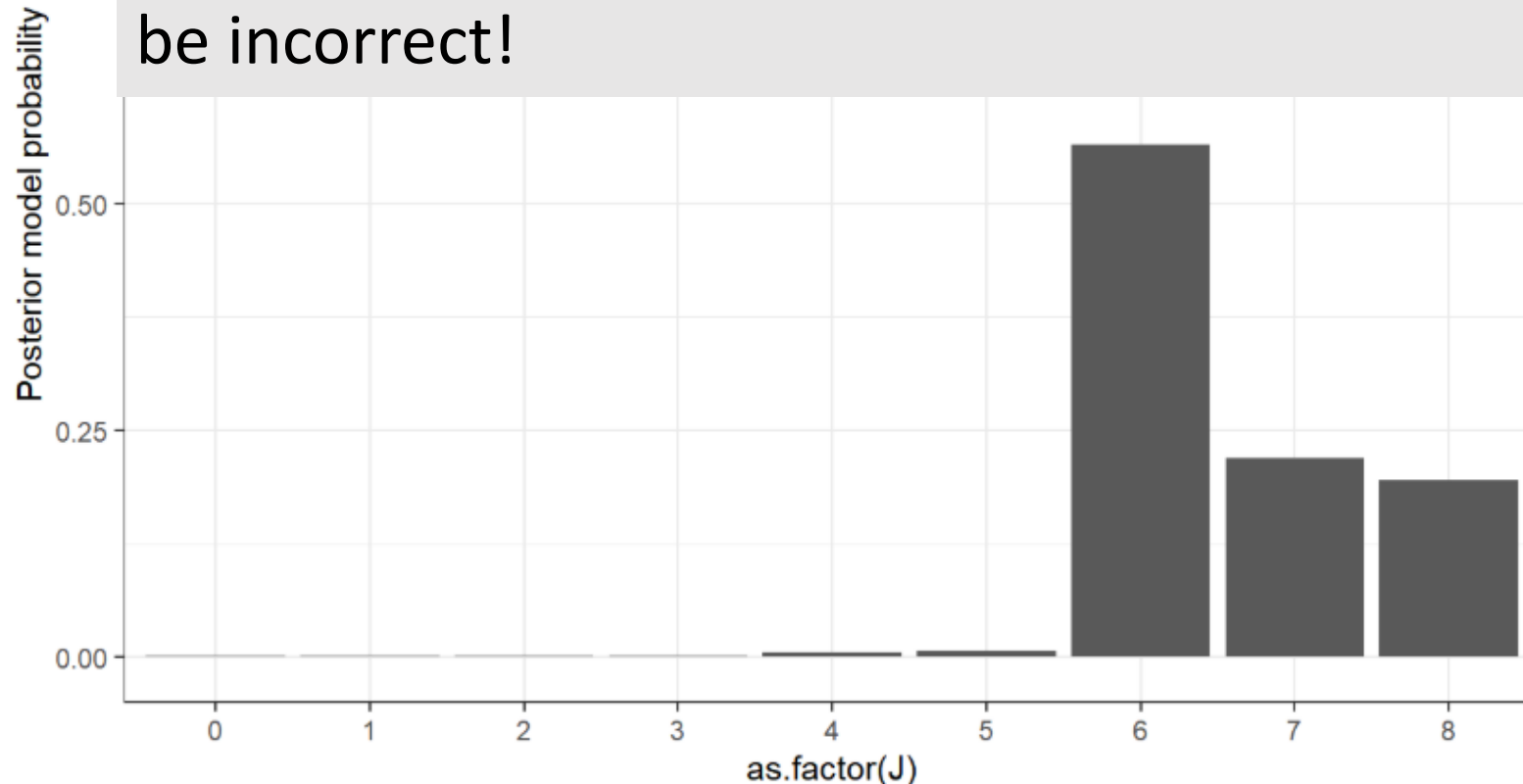
Will the log-evidence still identify the correct model?



Will the log-evidence still identify the correct model?

The 6th degree model is now considered to be the best!
Followed by the 7th and 8th degree models!

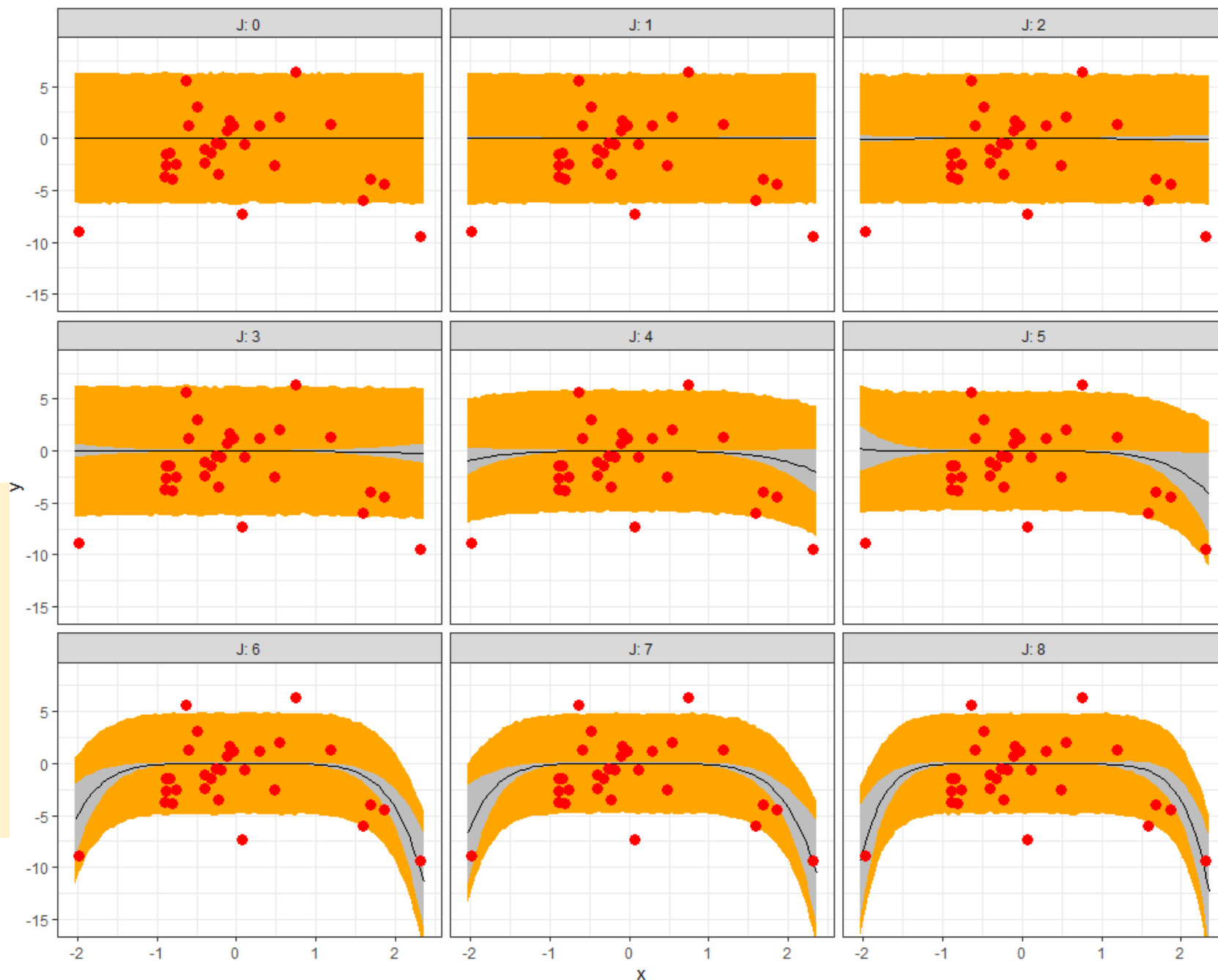
The quadratic model is now considered essentially to be incorrect!



Look at predictions from
ALL polynomials based on
the VERY STRONG prior

The quadratic model
behaves as a constant
without any trend!!

The very strong prior has
forced the parameter
values to be so close to
zero that only the highest
degree polynomials show
any trend!



Did we just break Bayes?

- By setting $\tau_\beta = 0.04$ we can no longer properly identify the correct model...

Did we just break Bayes?

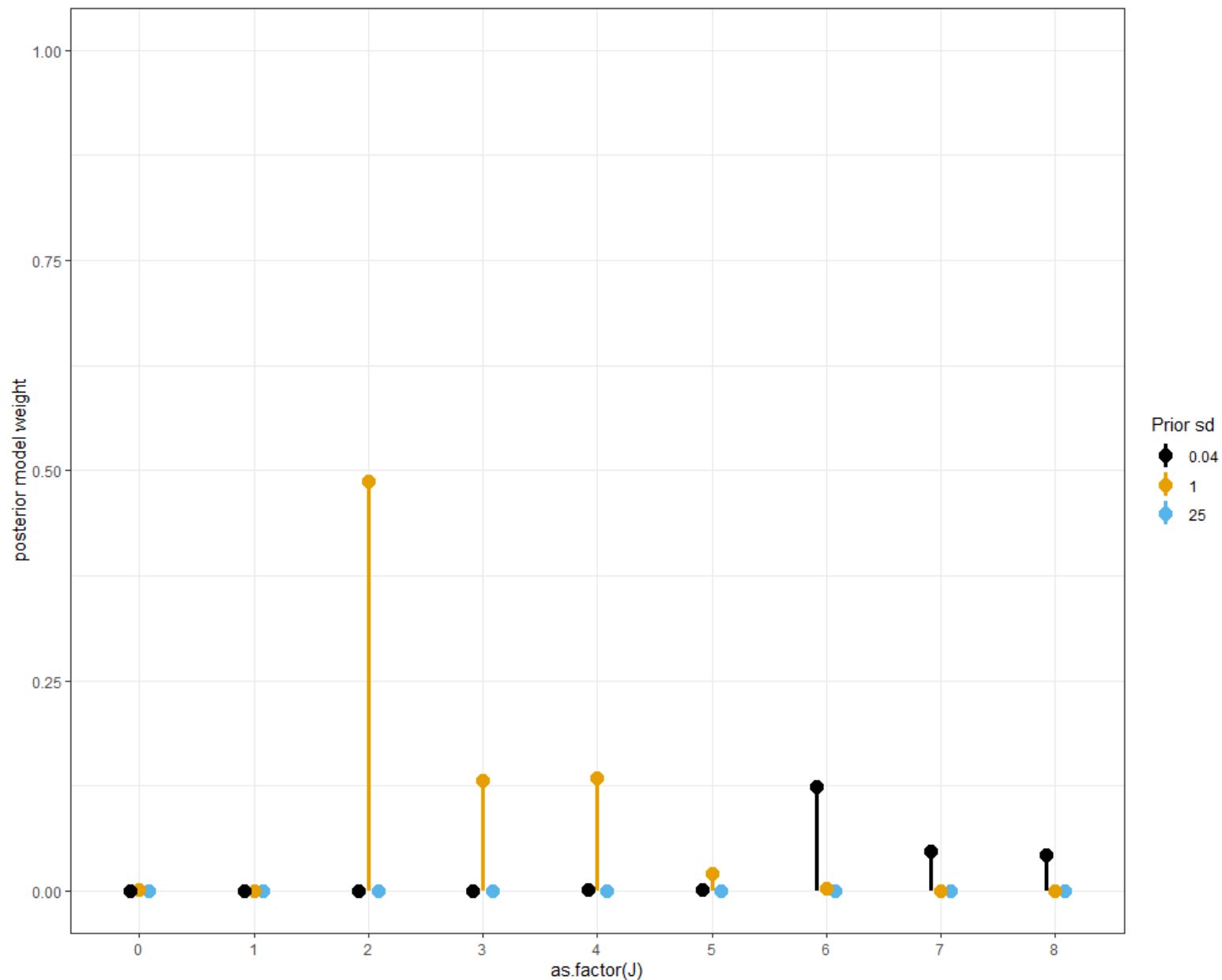
- By setting $\tau_\beta = 0.04$ we can no longer properly identify the correct model...
- No...

If we compare ALL models
with the posterior model
weight...

The quadratic model is still
the best BUT look closely!!

The quadratic model with the
STRONG prior, $\tau_\beta = 1$, is the
overall best!

So...Bayes still works as
expected...



But...if we had ONLY assumed the VERY STRONG prior...

- By setting $\tau_\beta = 0.04$ we can no longer properly identify the correct model...
- Is this wrong to do?

But...if we had ONLY assumed the VERY STRONG prior...

- By setting $\tau_\beta = 0.04$ we can no longer properly identify the correct model...
- Is this wrong to do?
- Technically no...we are allowed to do this...but should we?

Think about the model and the data!

- By specifying $\tau_\beta = 0.04$ we are saying that we believe that:
- The 95% prior uncertainty interval around the prior mean is:
$$-0.08 < \beta < 0.08.$$
- The 99.7% prior uncertainty interval around the prior mean is:
$$-0.12 < \beta < 0.12.$$

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- **With this prior, we're essentially saying there's no chance that a coefficient has a value of ± 0.25 , let alone a value of ± 1 .**

Unless you really think there should be NO trend between the response and the input...**I would NOT recommend a prior this *strong*.**

The β -parameters are slopes, they represent the change in the mean associated with a change in the input (feature)

- Consider the quadratic relationship: $\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,1}^2$
- If $\beta_1 = 1$, a change of 2 units of x induces a change of 2 units in the mean response.
- If $\beta_2 = 1$, a change in 2 units of x^2 induces a change of 2 units in the mean response. (2 unit change in x^2 is a $\sqrt{2}$ unit change in x).

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- A prior standard deviation of $\tau_\beta = 1$ says such slopes are “allowable”.
- However, with $\tau_\beta = 0.04$ such slopes are NOT “allowable” and are considered very unlikely.

If we have 3 separate inputs x_1 , x_2 , and x_3 ...

- With mean trend: $\mu_n = \beta_0 + \beta_1 x_{n,1} + \beta_2 x_{n,2} + \beta_3 x_{n,3}$.
- “Allowable” ranges on the slopes depend on the scale of the inputs AND on the scale of the response!

For example if I want to predict the distance of a homerun in baseball...

- I could consider the following inputs:
 - ambient (outside) air Pressure
 - The ball's "launch" speed
 - The wind speed
- The air Pressure is ≈ 100000 Pa
- The ball speed is about 100 mph
- The wind speed is around 5 mph

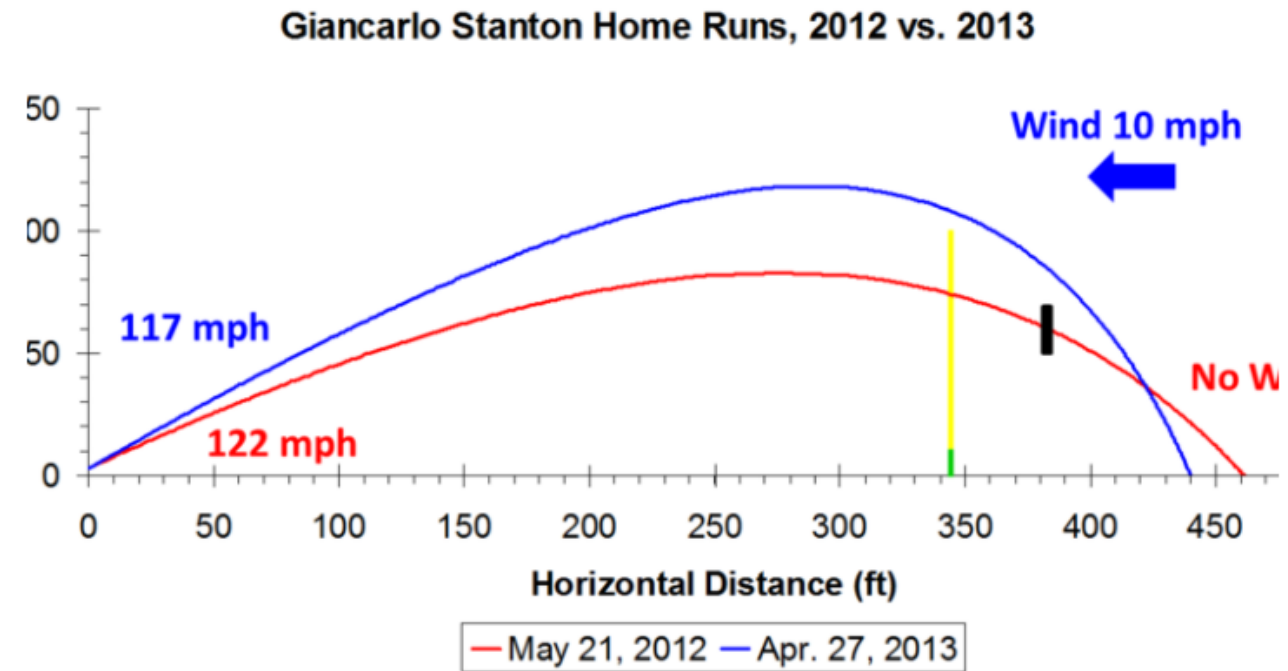


Image from: <https://www.sun-sentinel.com/sports/fl-xpm-2013-04-28-sfl-miami-marlins-giancarlo-stanton-home-run-distances-explained-20130428-story.html>

For example if I want to predict the distance of a homerun in baseball...

- The air Pressure is ≈ 100000 Pa
- The ball speed is about 100 mph
- The wind speed is around 5 mph
- A “strong” prior relative to the wind speed may not have any influence on the air pressure!
- A 1 unit change in wind speed may matter a lot more than 1 unit change in air Pressure!

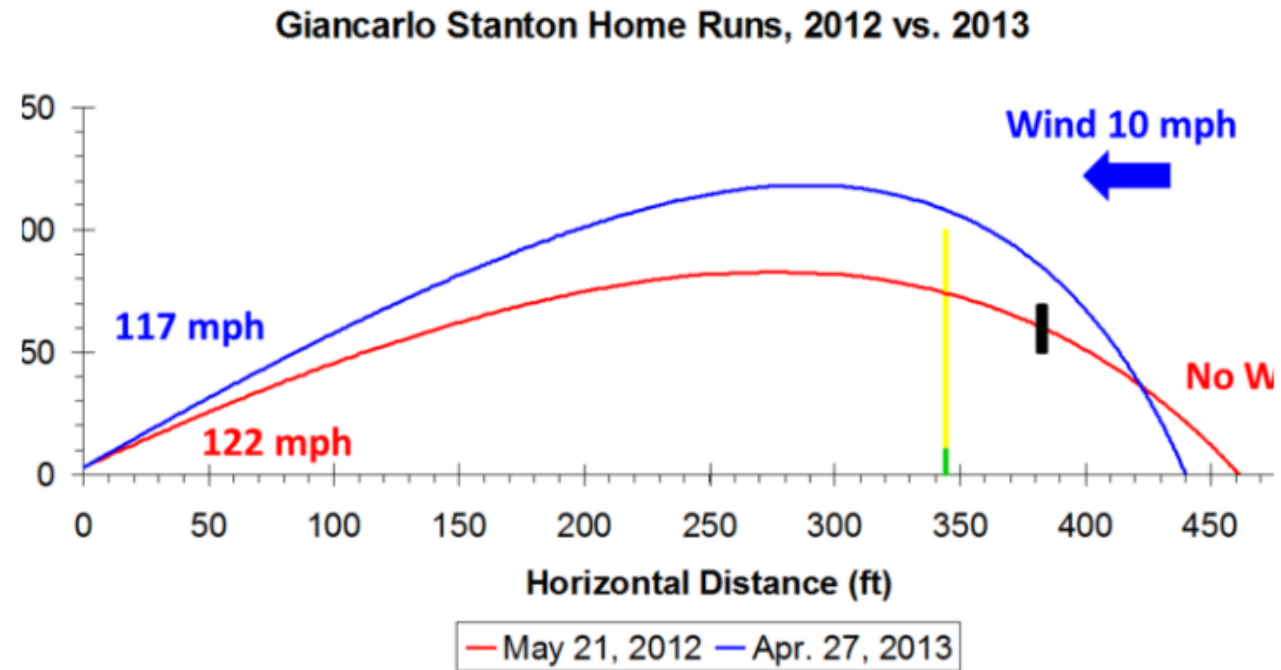


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Standardizing variables helps with establishing common prior specifications

- Rather than working with the “raw” input features and “raw” response we CENTER and SCALE such that all have comparable ranges.

- CENTER: subtract the sample average

$$x_{n,d}^{(\text{center})} = x_{n,d}^{(\text{raw})} - \text{mean}(\mathbf{x}_{:,d}^{(\text{raw})})$$

$$y_n^{(\text{center})} = y_n^{(\text{raw})} - \text{mean}(\mathbf{y}^{(\text{raw})})$$

- SCALE: divide by the sample standard deviation

$$x_{n,d} = \frac{x_{n,d}^{(\text{center})}}{\text{sd}(\mathbf{x}_{:,d}^{(\text{raw})})}, y_n = \frac{y_n^{(\text{center})}}{\text{sd}(\mathbf{y}^{(\text{raw})})}$$

Interpretations of the slopes are now on the STANDARDIZED scale

- If $\beta_1 = 1$, a change of 2 standardized units of x_1 induces a change of 2 standardized units in the mean response.
- If $\beta_2 = 1$, a change in 2 standardized units of x_2 induces a change of 2 standardized units in the mean response.

“Strong” and “weak” priors in the standardized scale

- A prior standard deviation of $\tau_\beta = 0.1$ allows changes of ± 2 standardized input units to cause a ± 0.4 standardized mean response units.
- A prior standard deviation of $\tau_\beta = 1$ allows changes of ± 2 standardized input units to cause a ± 4 standardized mean response units.
- A prior standard deviation of $\tau_\beta = 10$ allows changes of ± 2 standardized input units to cause a ± 40 standardized mean response units.

“Strong” and “weak” priors in the standardized scale

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- A prior standard deviation of $\tau_\beta = 1$ allows changes of ± 2 standardized input units to cause a ± 4 standardized mean response units.
- **A prior standard deviation of $\tau_\beta = 10$ allows changes of ± 2 standardized input units to cause a ± 40 standardized mean response units.**

40 standardized units corresponds to the response changing by more than 40 times the sample standard deviation around the sample average!!!!

Common or default priors used in practice

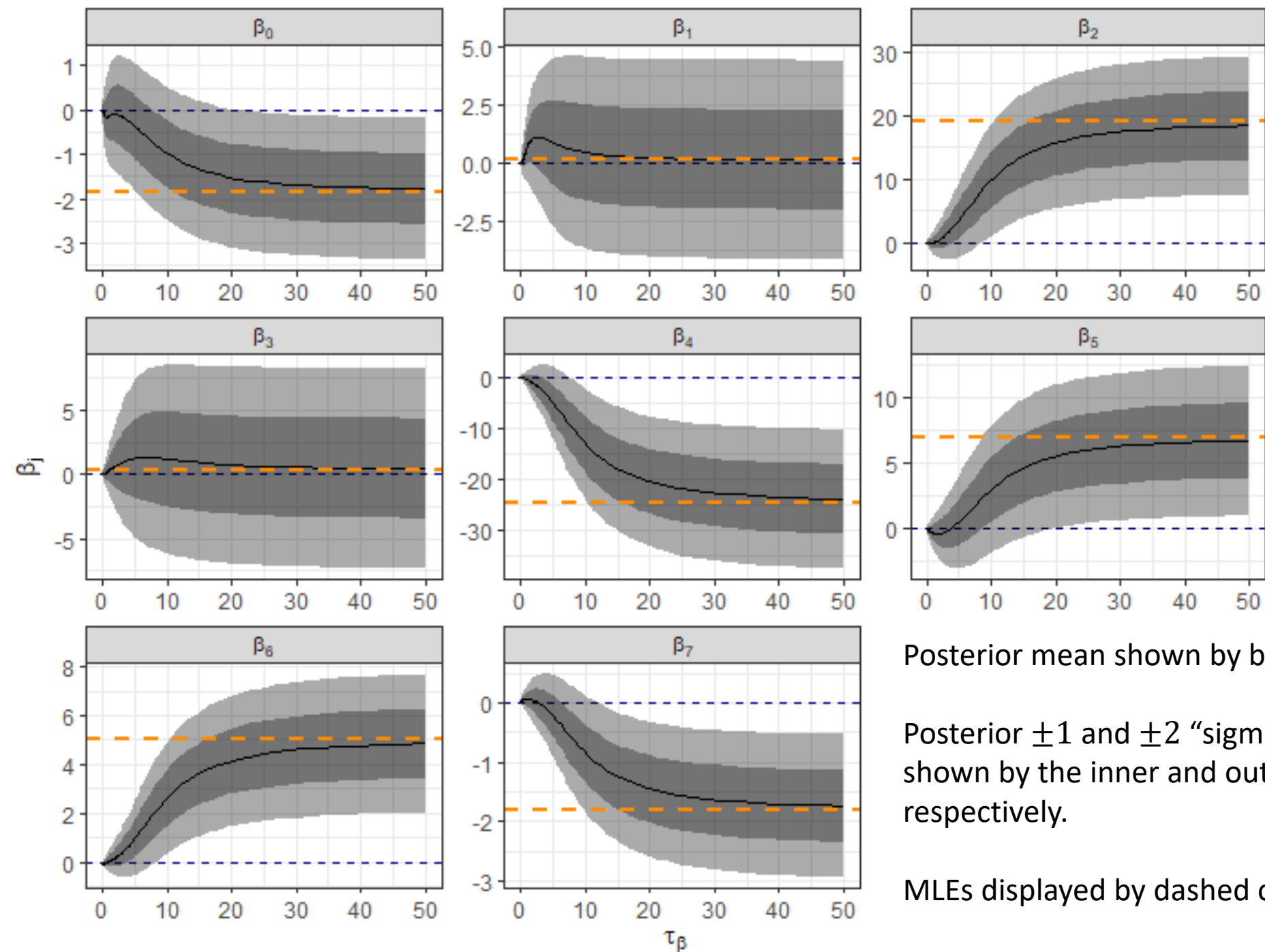
- Modern Bayesian practice for common or default priors is to use “weak” priors in the **standardized scale**.
 - τ_β between 1 to 10, with 3-5 typically used.
- **Very weak, infinitely diffuse, or “flat” priors are not suggested.**
- Very strong priors, $\tau_\beta < 1$, would only be recommended if you feel strongly that the effect should not matter!

Standardizing the output also helps specify the prior on σ

- A model that cannot explain any trends (an unknown constant average or intercept-only model) would have σ equal to the output's standard deviation.
- In the standardized scale, the output's standard deviation is 1!!!
- Thus, if we assume a-priori the expected σ is 1 in the standardized scale...we are seeing if the trend can do better than an unknown constant average!!!!

Let's get a more detailed view by varying τ_β from 0.02 to 50

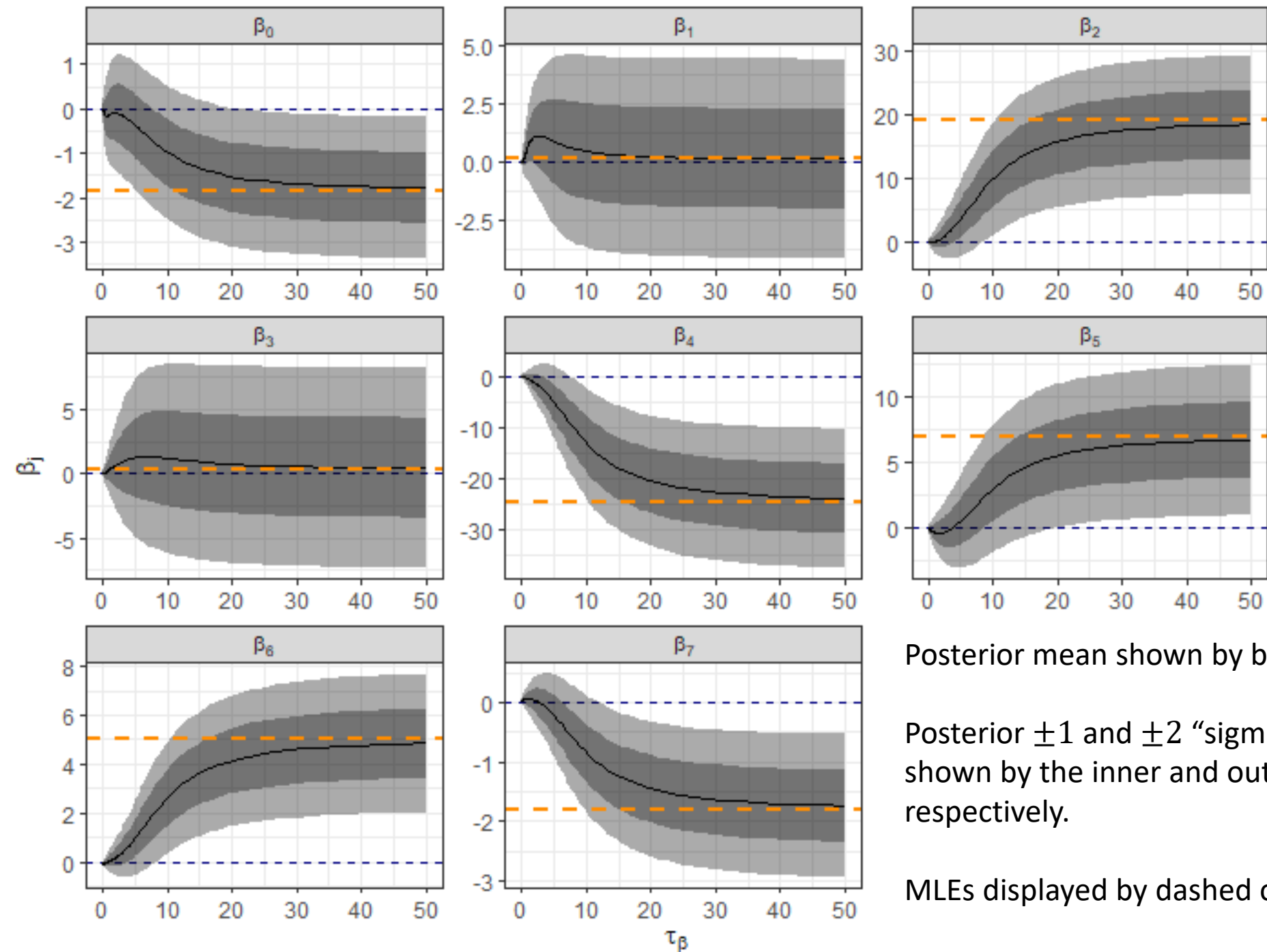
- First, consider the 7th order polynomial.
- Let's “watch” what happens to the coefficients as we increase the prior standard deviation.



Posterior mean shown by black line.

Posterior ± 1 and ± 2 “sigma” uncertainty intervals shown by the inner and outer grey ribbons, respectively.

MLEs displayed by dashed orange lines for reference.



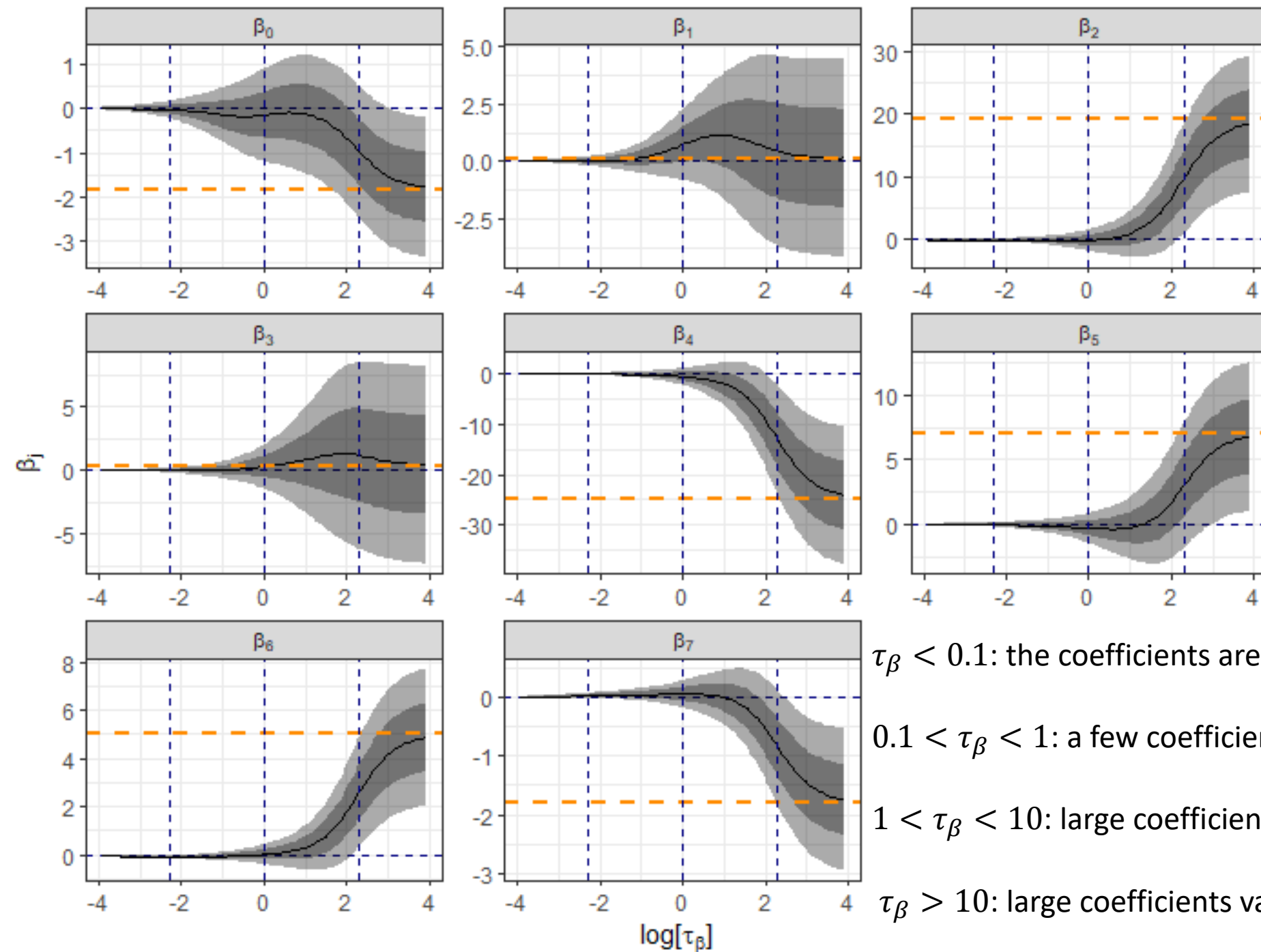
Posterior mean shown by black line.

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MLEs displayed by dashed orange lines for reference.

Plot the coefficients with respect to $\log[\tau_\beta]$ to help see what's happening for the small very informative priors

- The following figure displays dashed vertical lines at $\log[\tau_\beta]$ equal to $\log[0.1]$, $\log[1]$, and $\log[10]$ for reference.



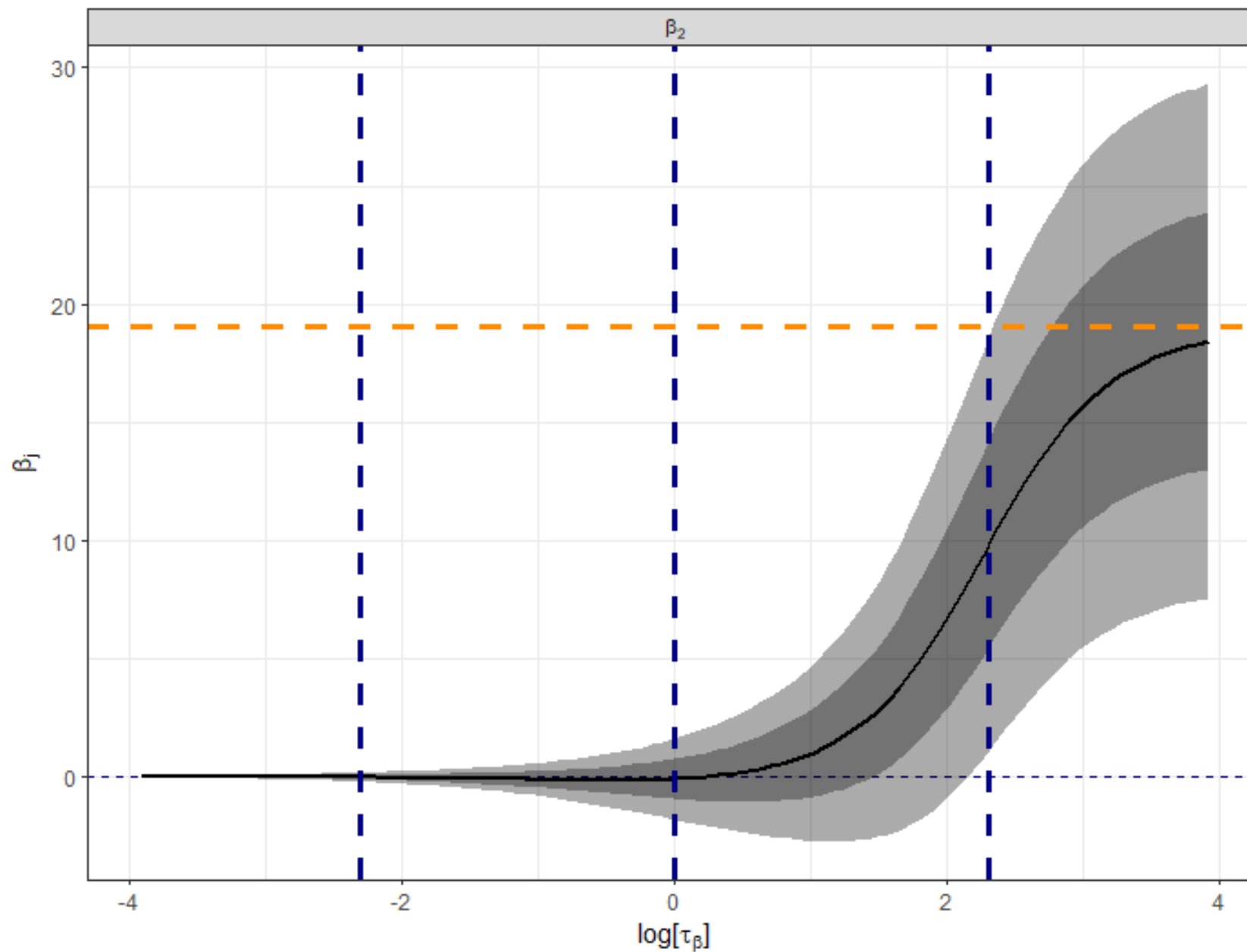
$\tau_\beta < 0.1$: the coefficients are all close to zero, the prior mean.

$0.1 < \tau_\beta < 1$: a few coefficients move away from zero.

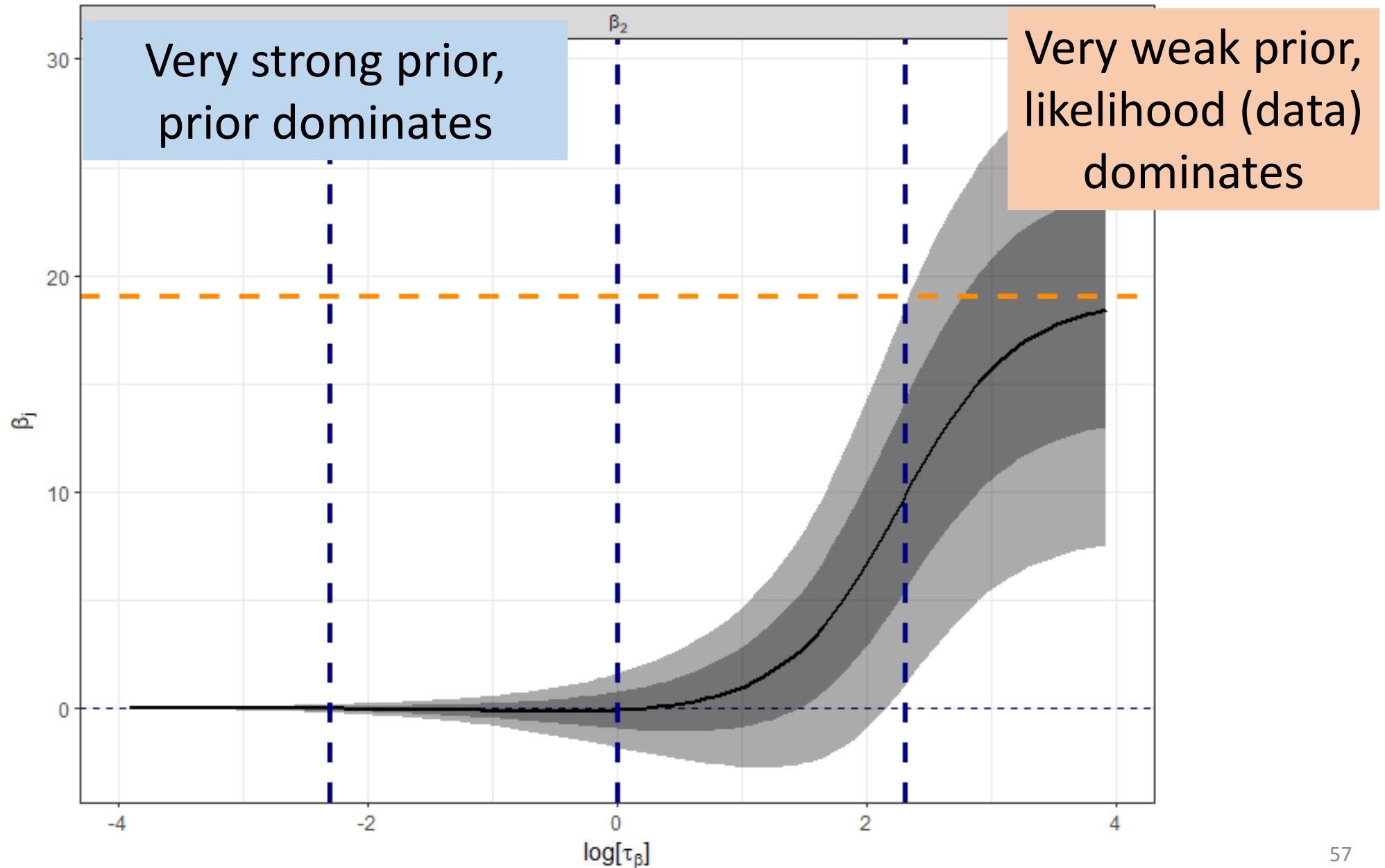
$1 < \tau_\beta < 10$: large coefficient values are prevented.

$\tau_\beta > 10$: large coefficients values are allowed, converge to MLEs.

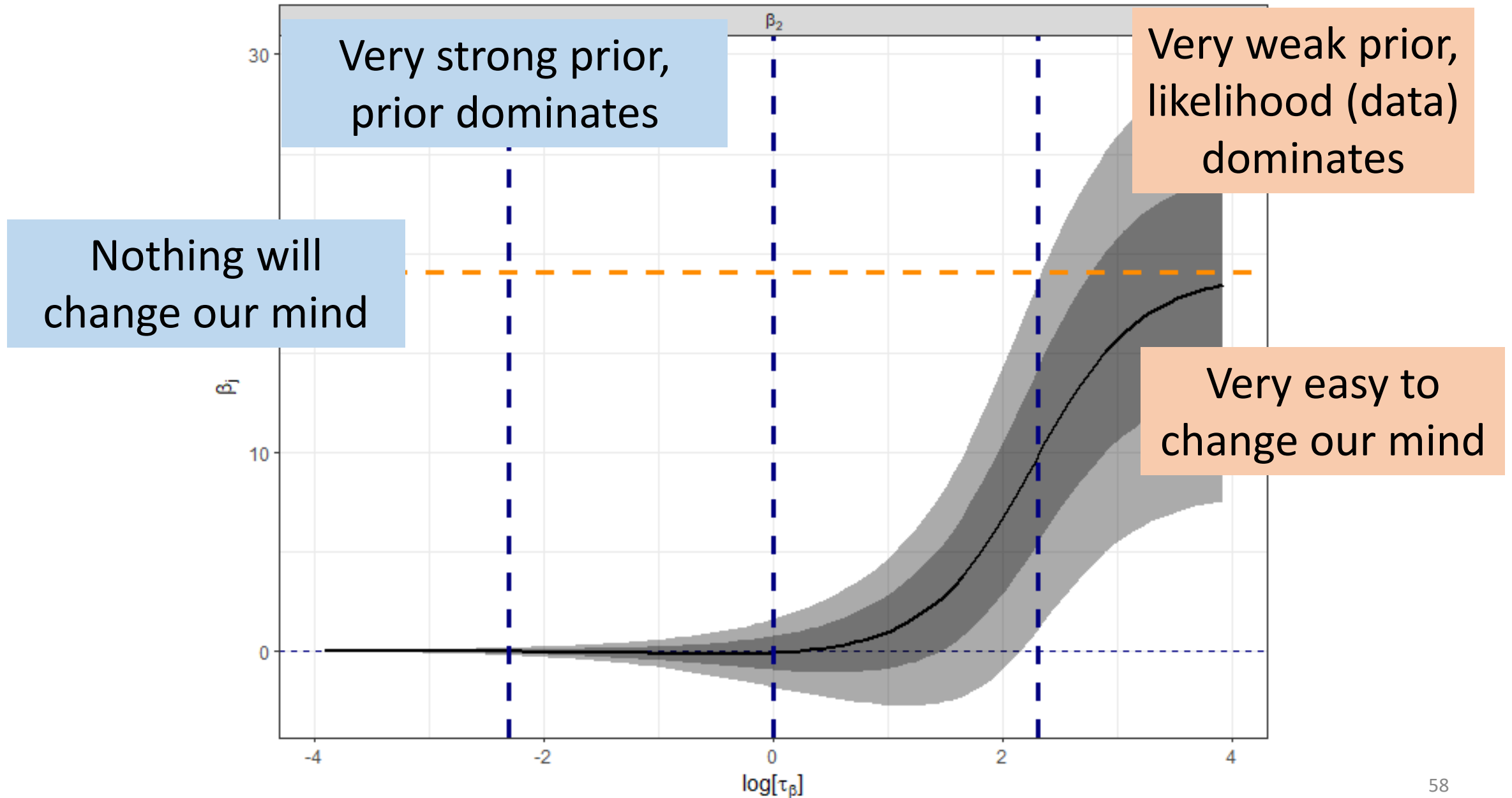
Focus just on β_2 , the slope of the quadratic term in the 7th order polynomial



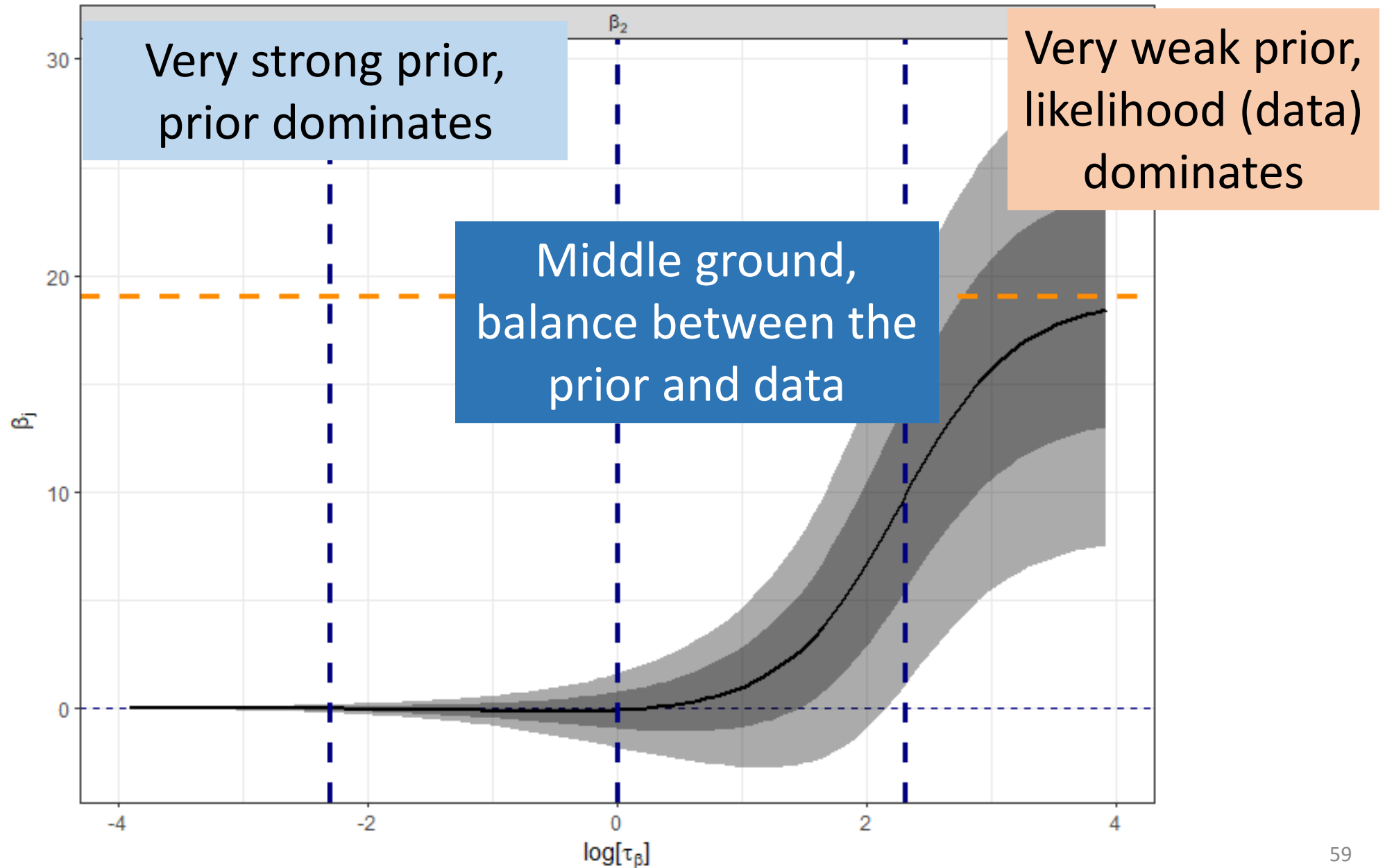
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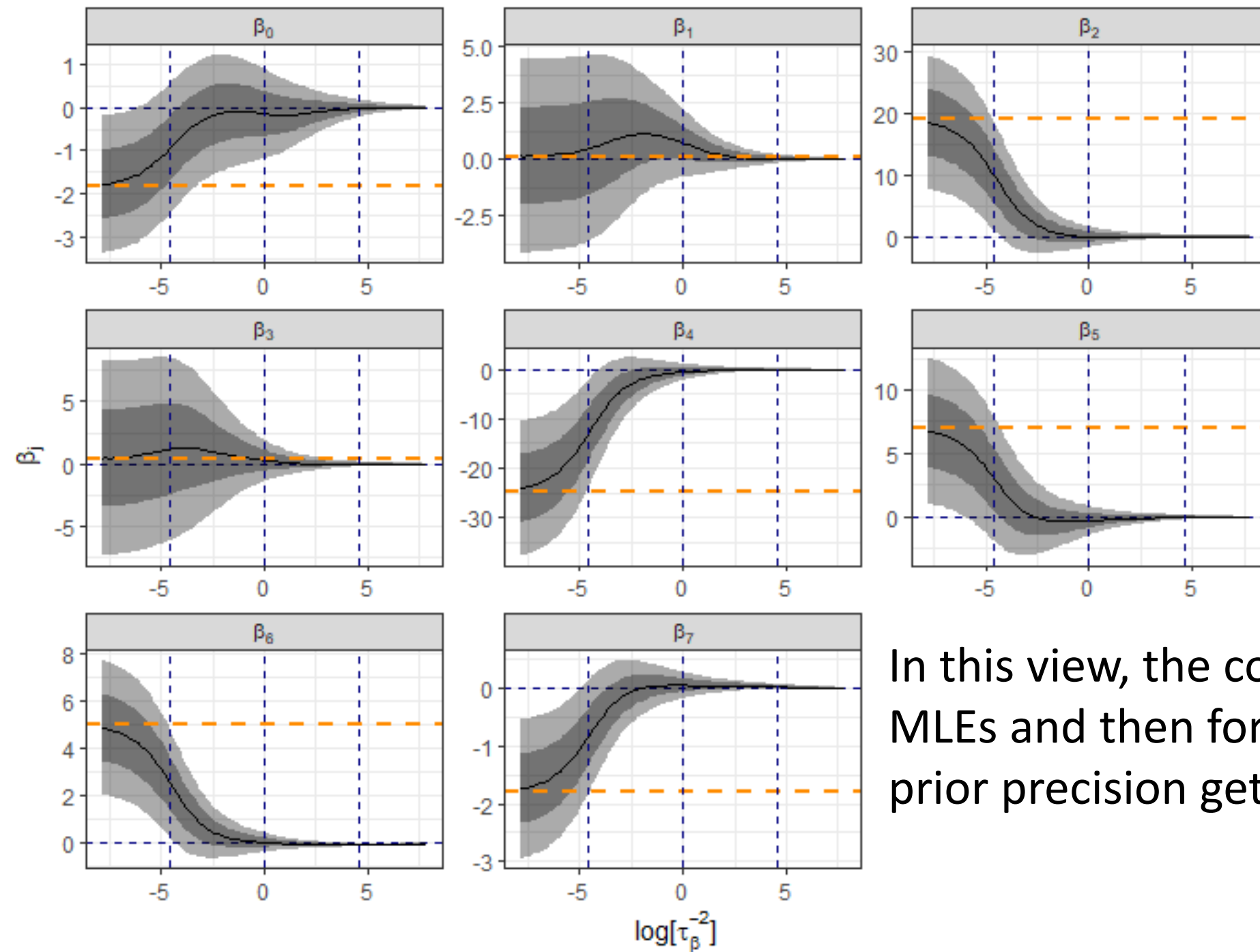


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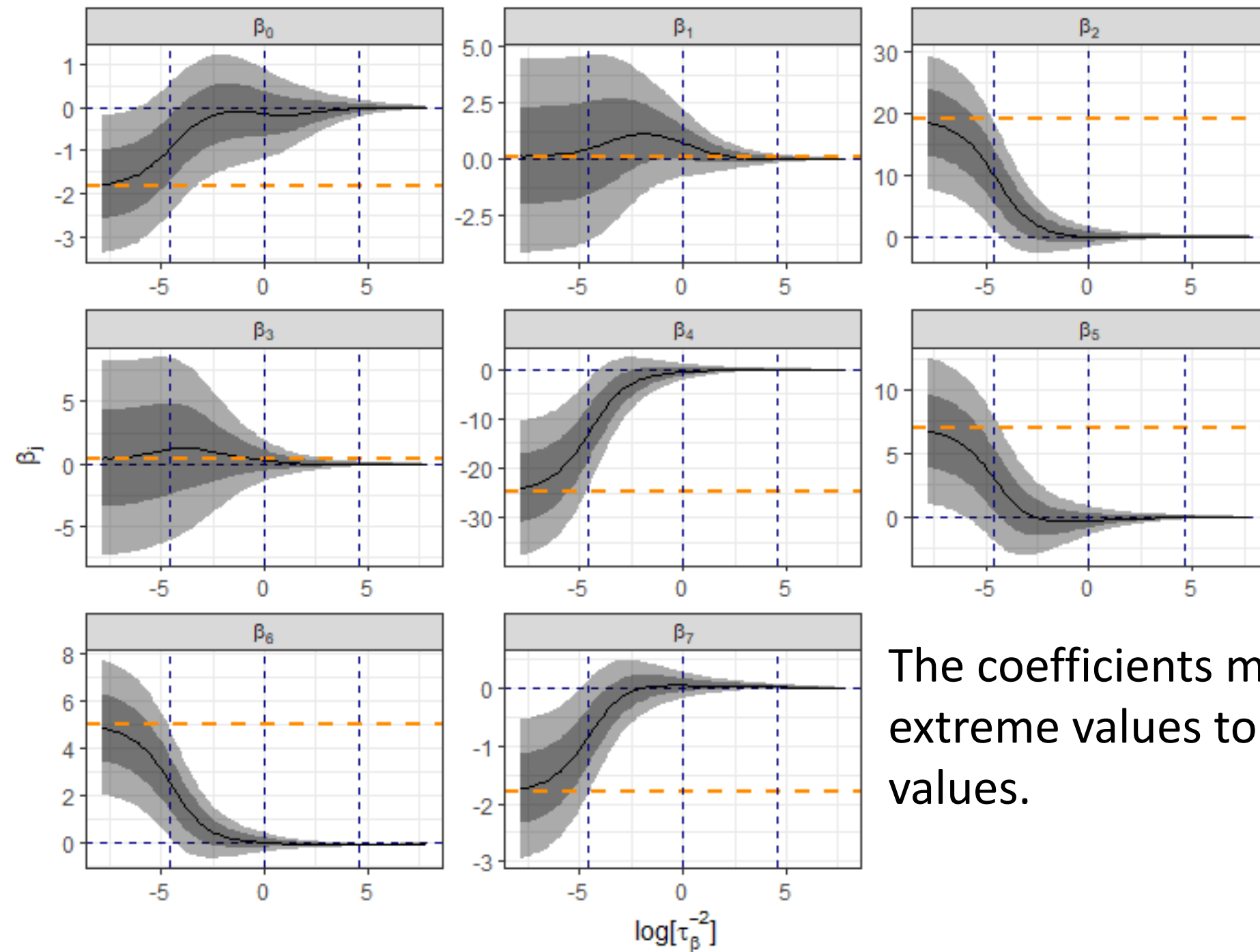


For historical purposes, plot the coefficients with respect to the log of the prior precision

- Plotting β_j vs $\log[\tau_\beta^{-2}]$ shows the coefficients forced to zero as the prior precision increases.



In this view, the coefficients start at their MLEs and then forced to zero as the prior precision gets higher and higher.



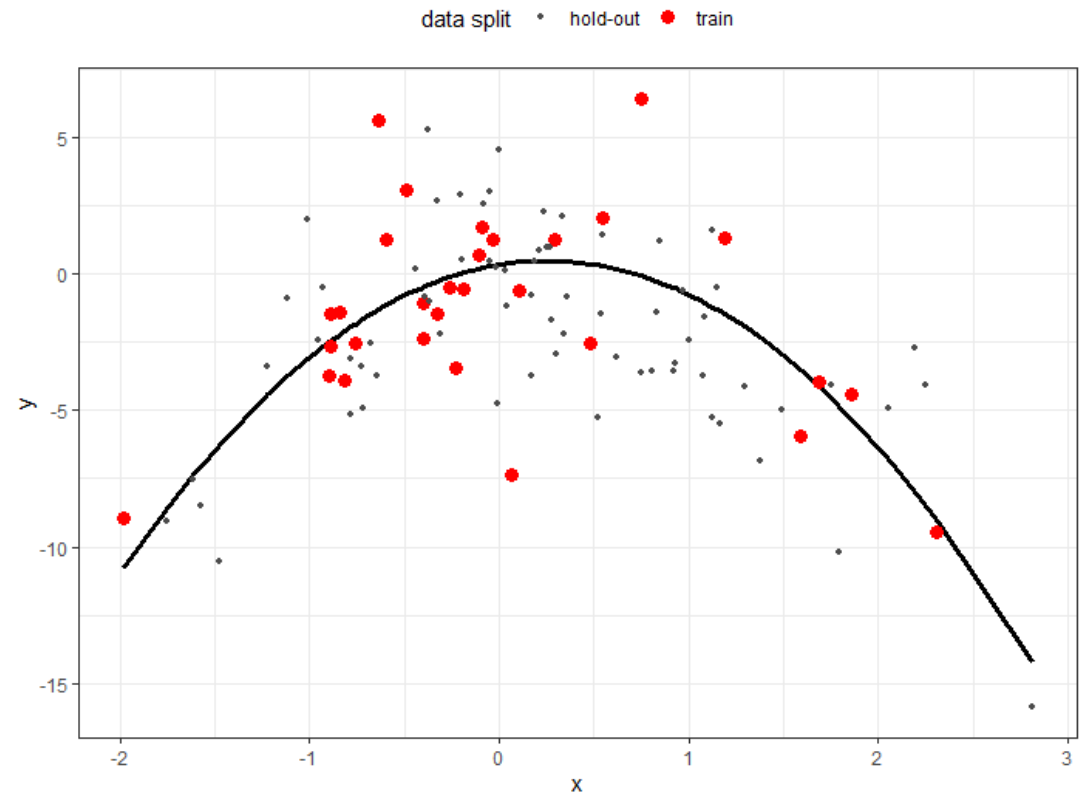
The coefficients move from potentially extreme values to more “regular” values.

Constraining coefficients away from large values is known as **REGULARIZATION**

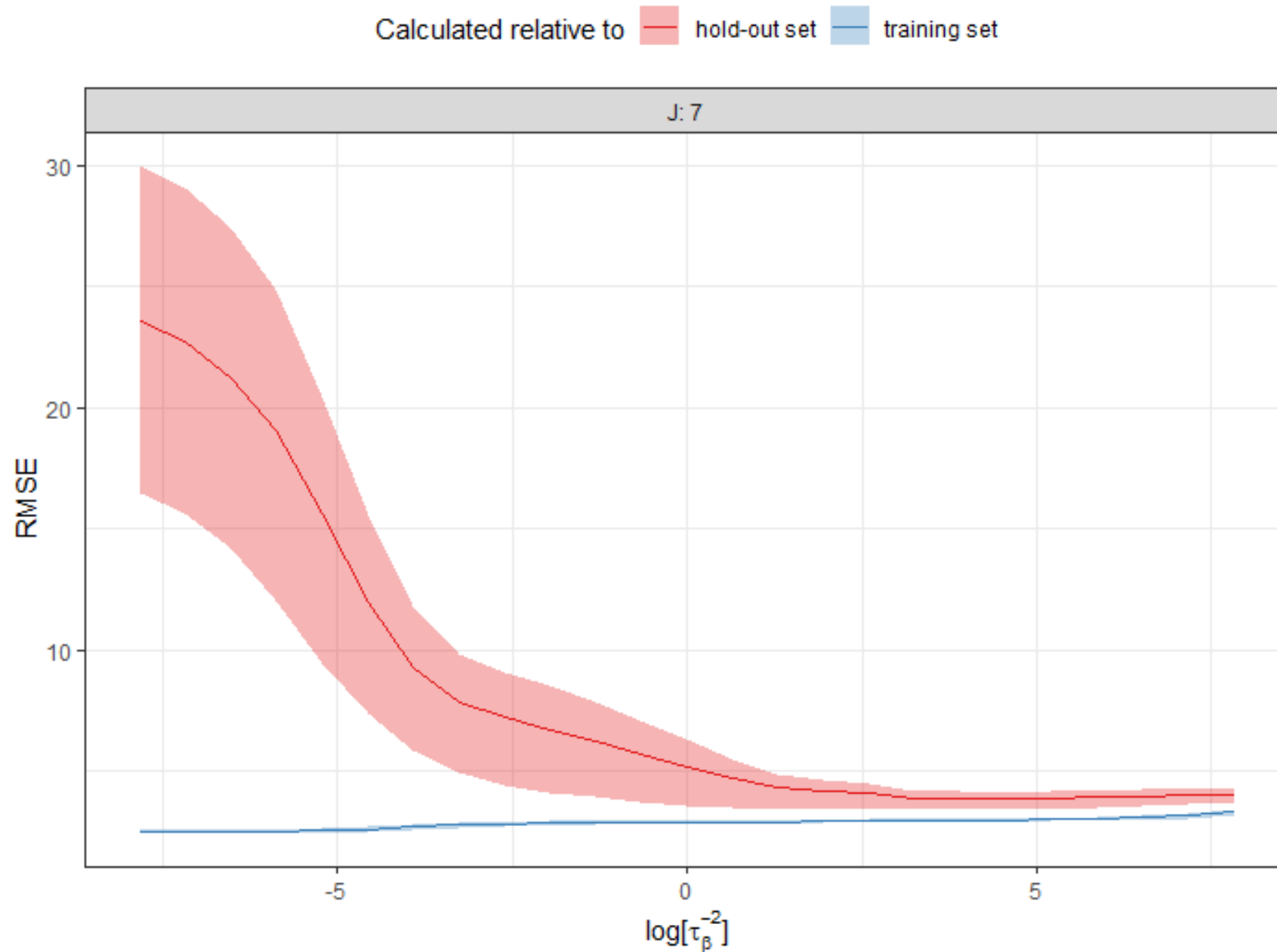
- The priors regularize the coefficients!

Let's see the effect of the regularization on the predictive performance

- 100 data points were generated.
- 30 were selected as the training set.
- Use the remaining 70 as a hold-out test set.
- Calculate RMSE on the training set and hold-out set with respect to τ_β

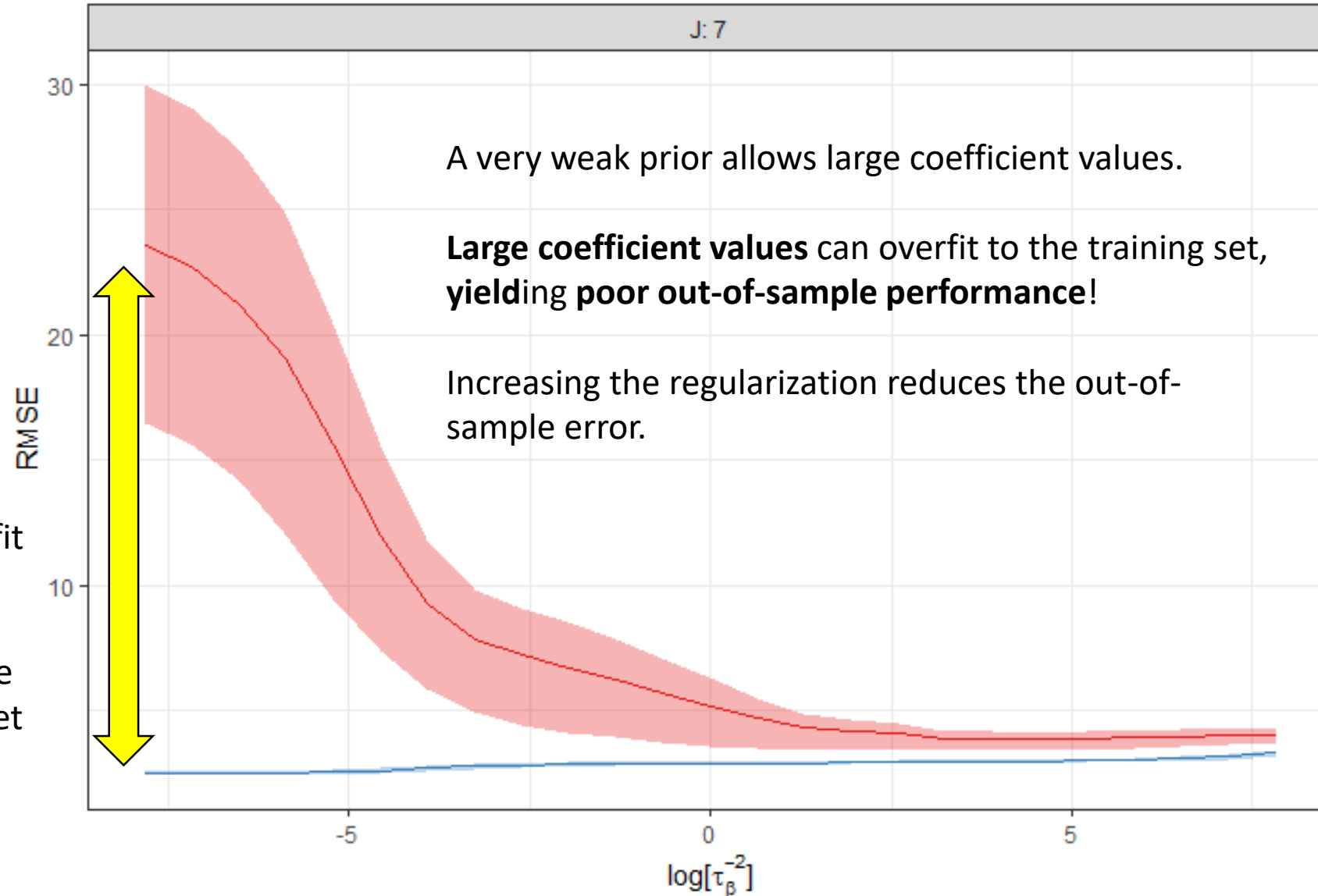


RMSE vs the log-prior precision for the 7th degree polynomial



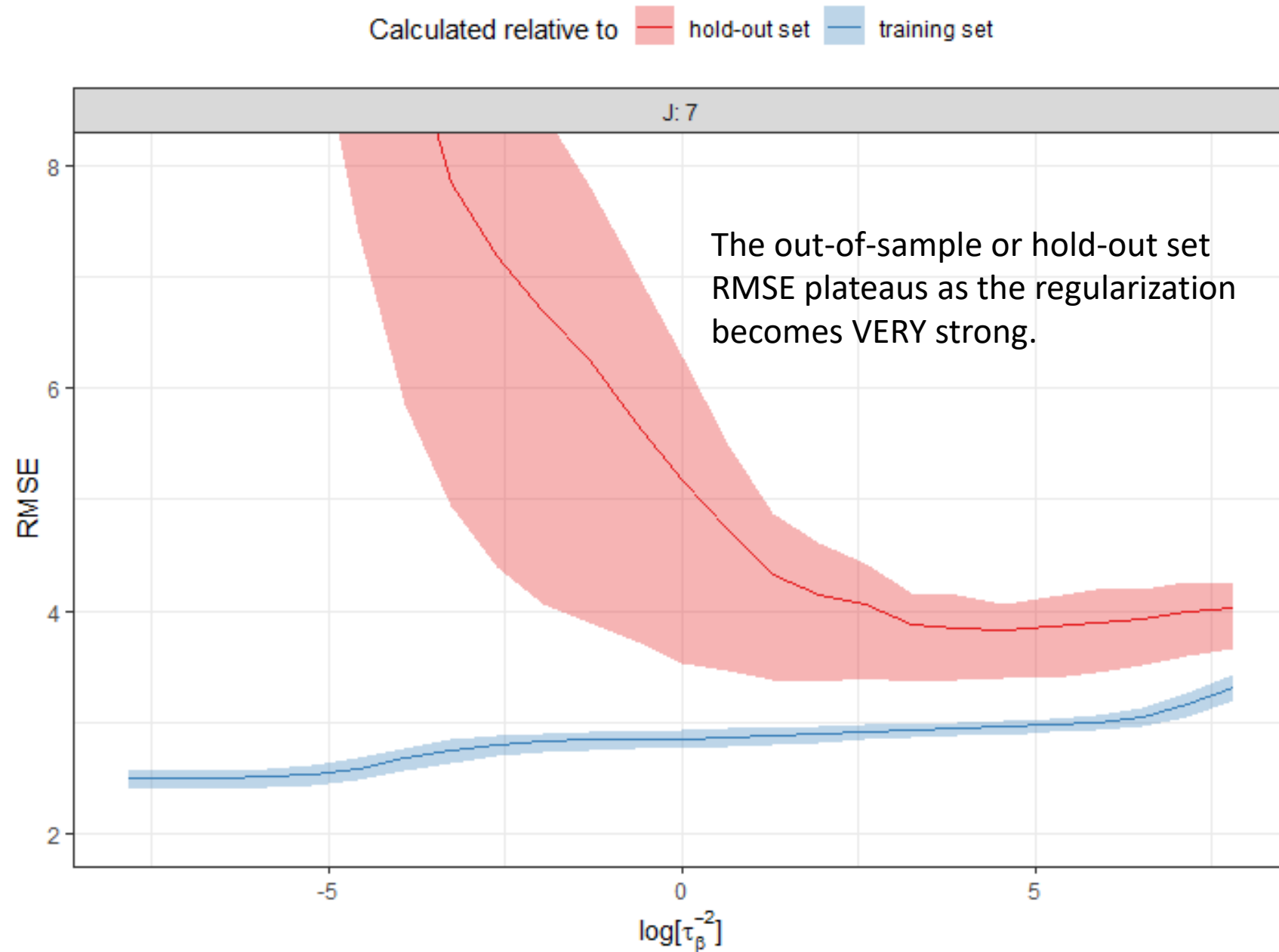
RMSE vs the log-prior precision for the 7th degree polynomial

Calculated relative to — hold-out set — training set

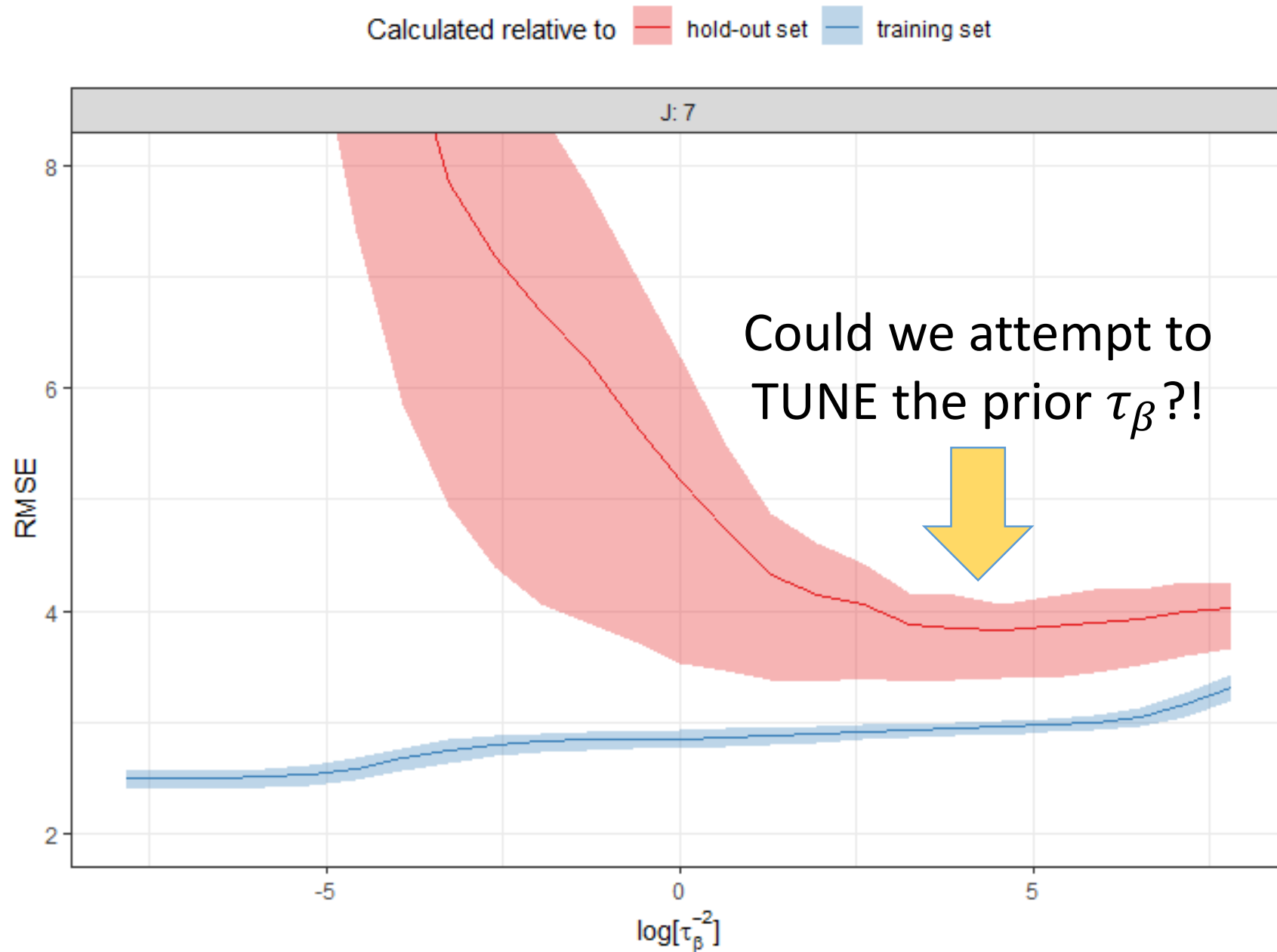


We know it's overfit because of the **large** difference in errors between the training and test set

Zoom in on the RMSE vs the log-prior precision for the 7th degree polynomial



Zoom in on the RMSE vs the log-prior precision for the 7th degree polynomial



Regularization can be viewed as limiting or even *controlling* overfitting!

- The prior standard deviation sets the regularization, and so controls the amount of overfitting.
- **We can start from a quite complex model and “turn it” into a simpler model.**
- But what about the impact of the noise, σ ?

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- We have used independent Gaussian priors with a shared prior mean and shared prior standard deviation:

$$\boldsymbol{\beta} \mid \mu_{\beta}, \tau_{\beta} \sim \prod_{j=0}^J \left(\text{normal}(\beta_j \mid \mu_{\beta}, \tau_{\beta}) \right)$$

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- The independent log-prior summation is proportional to:

$$\log[p(\boldsymbol{\beta} \mid \mu_{\beta}, \tau_{\beta})] \propto -\frac{1}{2\tau_{\beta}^2} \sum_{j=0}^J \left((\beta_j - \mu_{\beta})^2 \right)$$

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- Set the prior mean to zero:

$$\log[p(\boldsymbol{\beta} \mid \mu_{\beta} = 0, \tau_{\beta})] \propto -\frac{1}{2\tau_{\beta}^2} \sum_{j=0}^J \left((\beta_j - 0)^2 \right) = -\frac{1}{2\tau_{\beta}^2} \sum_{j=0}^J (\beta_j^2)$$

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- Substitute in the prior's contribution to the un-normalized log-posterior:

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] - \frac{1}{2\tau_{\beta}^2} \sum_{j=0}^J (\beta_j^2)$$

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- The log-likelihood is proportional to the negative of the SSE!

$$SSE = \sum_{n=1}^N \left((y_n - \mathbf{x}_{n,:} \boldsymbol{\beta})^2 \right) = (\mathbf{y} - \mathbf{X} \boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X} \boldsymbol{\beta})$$



$$\log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] \propto -\frac{1}{2\sigma^2} SSE$$

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- Substitute the log-likelihood's contribution to the log-posterior:

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto -\frac{1}{2\sigma^2} SSE - \frac{1}{2\tau_{\boldsymbol{\beta}}^2} \sum_{j=0}^J (\beta_j^2)$$

The un-normalized log-posterior on the $\boldsymbol{\beta}$ -parameters assuming a known σ

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto \log[p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma)] + \log[p(\boldsymbol{\beta})]$$

- The log-posterior is a weighted balance between the error and the constraint from the prior!

$$\log[p(\boldsymbol{\beta} \mid \mathbf{y}, \mathbf{X}, \sigma)] \propto -\frac{1}{2\sigma^2} SSE - \frac{1}{2\tau_{\beta}^2} \sum_{j=0}^J (\beta_j^2)$$

Rearrange by factoring out the noise

$$-\frac{1}{2\sigma^2}SSE - \frac{1}{2\tau_\beta^2} \sum_{j=0}^J (\beta_j^2)$$



$$-\frac{1}{2\sigma^2} \left(SSE + \frac{\sigma^2}{\tau_\beta^2} \sum_{j=0}^J (\beta_j^2) \right)$$

Define the **regularization** or **penalization factor** λ

$$-\frac{1}{2\sigma^2} \left(SSE + \left(\frac{\sigma}{\tau_\beta} \right)^2 \sum_{j=0}^J (\beta_j^2) \right)$$

λ is used for historical purposes, this is NOT the prior rate on σ !

$$\lambda = \left(\frac{\sigma}{\tau_\beta} \right)^2$$

Focus on the expression within the parentheses

$$-\frac{1}{2\sigma^2} \left(SSE + \lambda \sum_{j=0}^J (\beta_j^2) \right)$$

Maximizing the likelihood is equivalent to minimizing the sum of squared errors (SSE)...

$$SSE + \lambda \sum_{j=0}^J (\beta_j^2)$$

...the prior is added to the SSE!

Maximizing the likelihood is equivalent to minimizing the sum of squared errors (SSE)...

$$SSE + \lambda \sum_{j=0}^J (\beta_j^2)$$

...the prior is added to the SSE! **The prior acts as a FLOOR or lower limit to minimizing the SSE!**

If large β values force the SSE to be very low, the prior's floor increases!

$$SSE + \lambda \sum_{j=0}^J (\beta_j^2)$$

The prior therefore pulls the coefficients away from large values!

When does the prior have no effect?

$$SSE + \lambda \sum_{j=0}^J (\beta_j^2)$$

If all β 's equal 0...but that means there are no TRENDS! So not entirely “interesting”...

When does the prior have no effect?

$$SSE + \lambda \sum_{j=0}^J (\beta_j^2)$$

The prior has no effect if the REGULARIZATION STRENGTH is ZERO!

When does the prior have no effect?

$$SSE + \lambda \sum_{j=0}^J (\beta_j^2)$$

When does $\lambda \rightarrow 0$? When $\tau_\beta \rightarrow \infty$!!!

Regularization also exists in non-Bayesian settings

$$\operatorname{argmin}_{\boldsymbol{\beta}} \left\{ SSE(\boldsymbol{\beta}) + \lambda \sum_{j=0}^J (\beta_j^2) \right\}$$

Referred to as regularized or penalized regression.

If the PENALTY term is equal to the sum of the squared coefficient values, referred to as RIDGE regression

$$\operatorname{argmin}_{\boldsymbol{\beta}} \left\{ SSE(\boldsymbol{\beta}) + \lambda \sum_{j=0}^J (\beta_j^2) \right\}$$

Non-Bayesian analysis typically tunes the penalty factor λ through cross-validation.

If the PENALTY term is equal to the sum of the absolute coefficient values, referred to as LASSO regression

$$\operatorname{argmin}_{\boldsymbol{\beta}} \left\{ SSE(\boldsymbol{\beta}) + \lambda \sum_{j=0}^J (|\beta_j|) \right\}$$

The penalty factor λ is tuned through cross-validation.

Ridge regression is the non-Bayesian analog of Bayesian models with independent Gaussian priors

- Lasso regression can also be interpreted from a Bayesian viewpoint.
- Requires introducing a new distribution – the Double Exponential or Laplace distribution.
- Visualizing the Double Exponential distribution can help provide context to the different behavior between ridge and lasso regression.

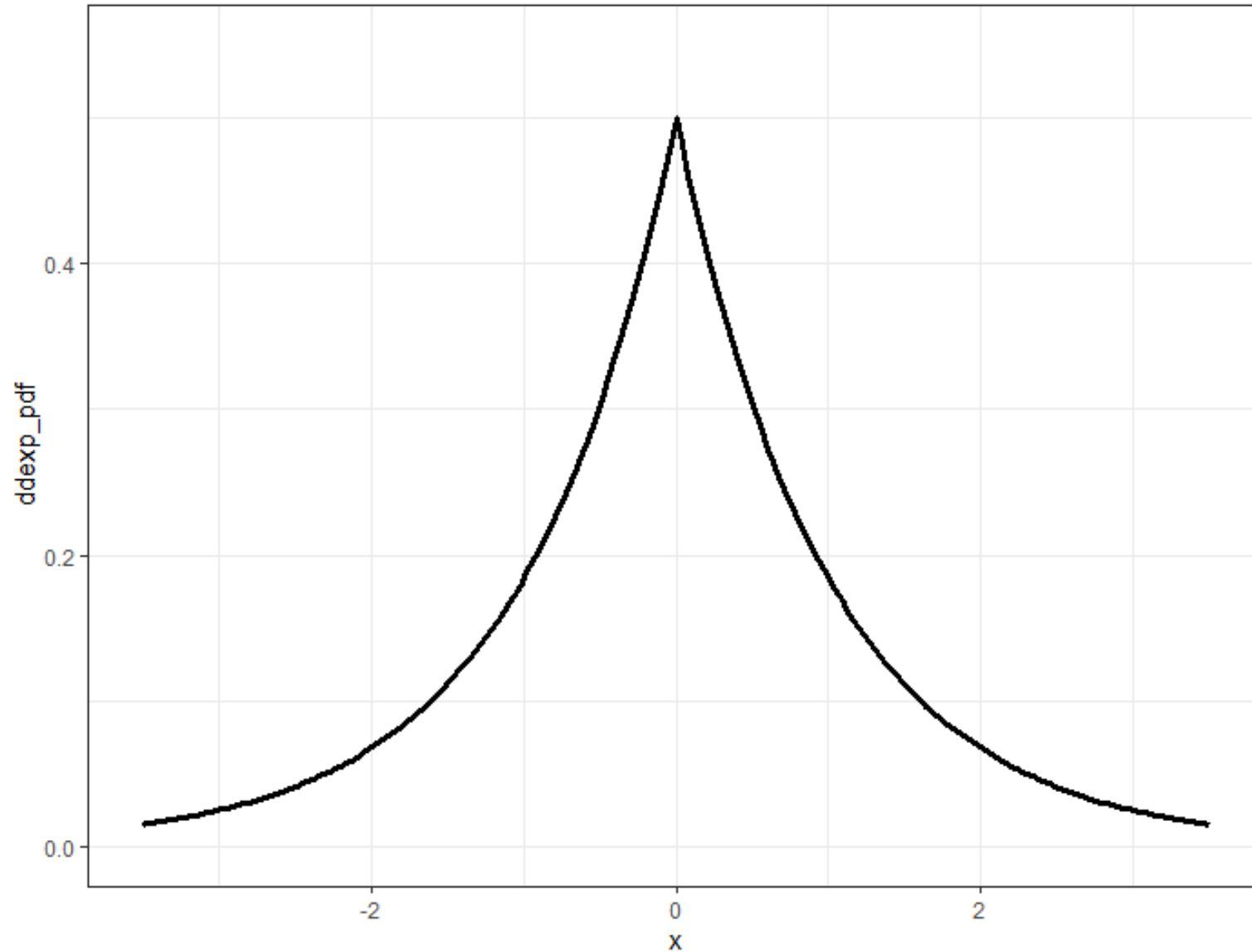
The Double Exponential pdf for a generic random variable x with location μ and scale b

$$x \mid \mu, b \sim \frac{1}{2b} \exp\left(-\frac{1}{b} |x - \mu|\right)$$

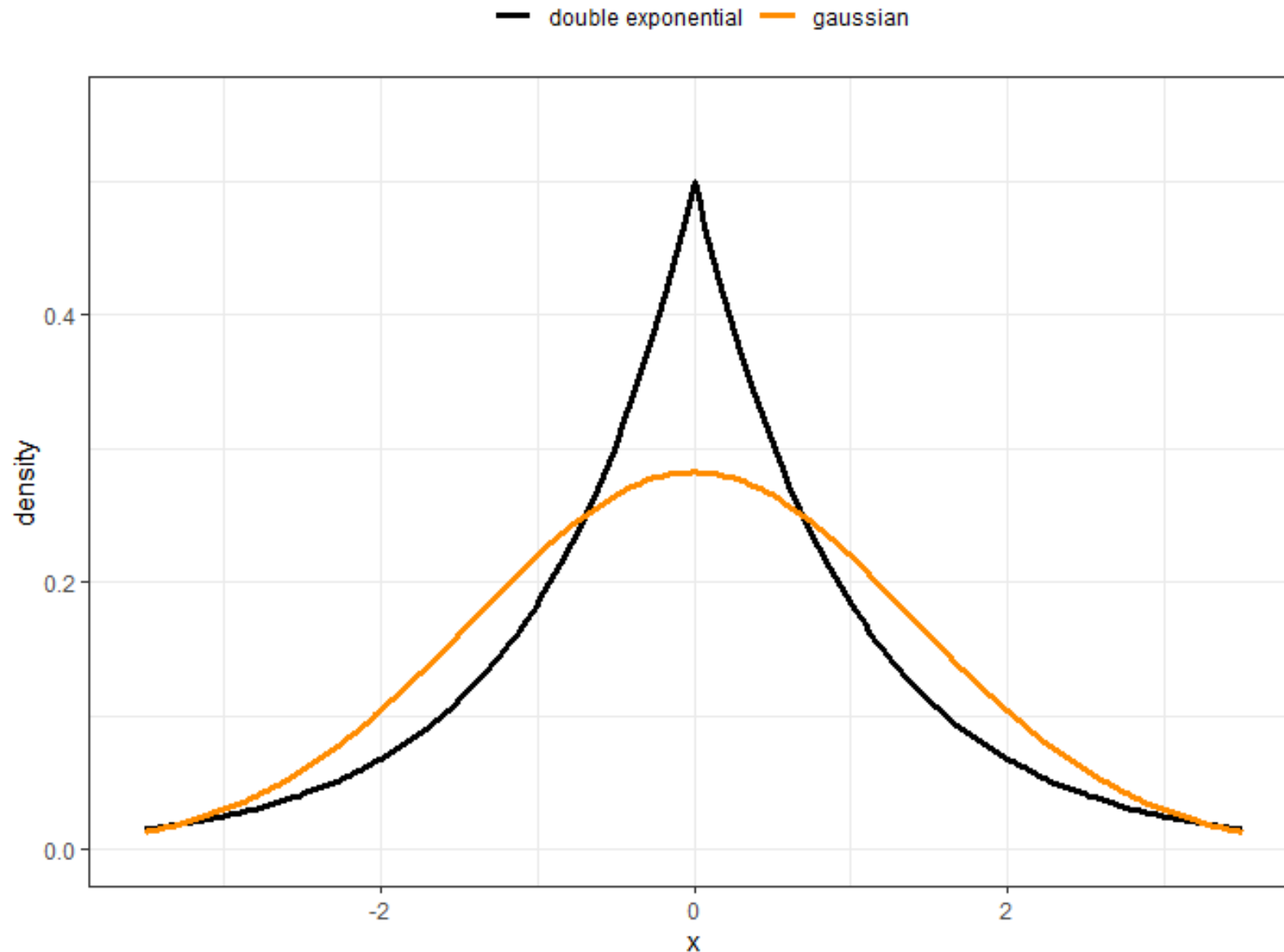
The expected value (mean): $\mathbb{E}[x \mid \mu, b] = \mu$.

The variance is: $\text{var}(x \mid \mu, b) = 2b^2$

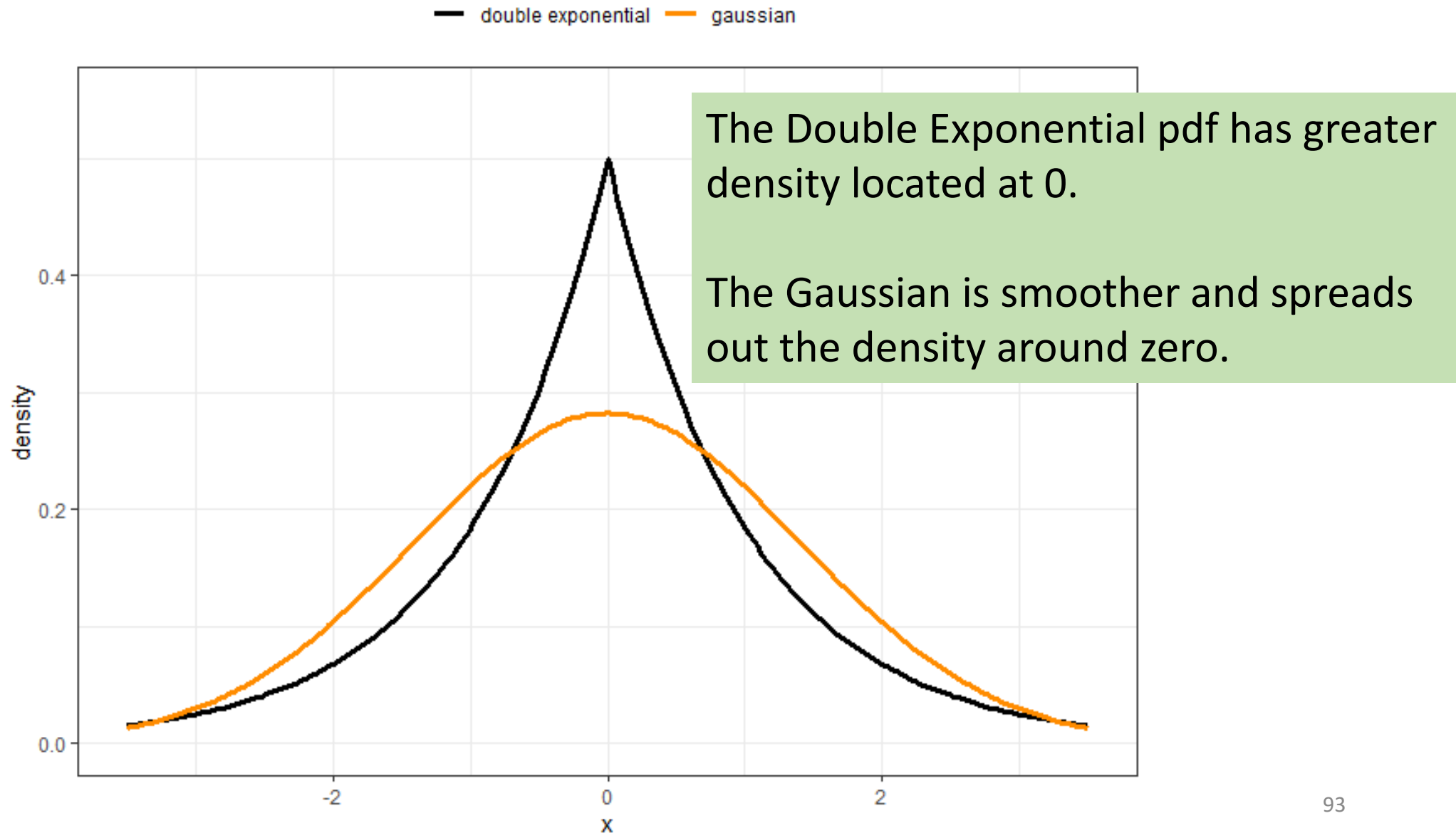
Double Exponential pdf



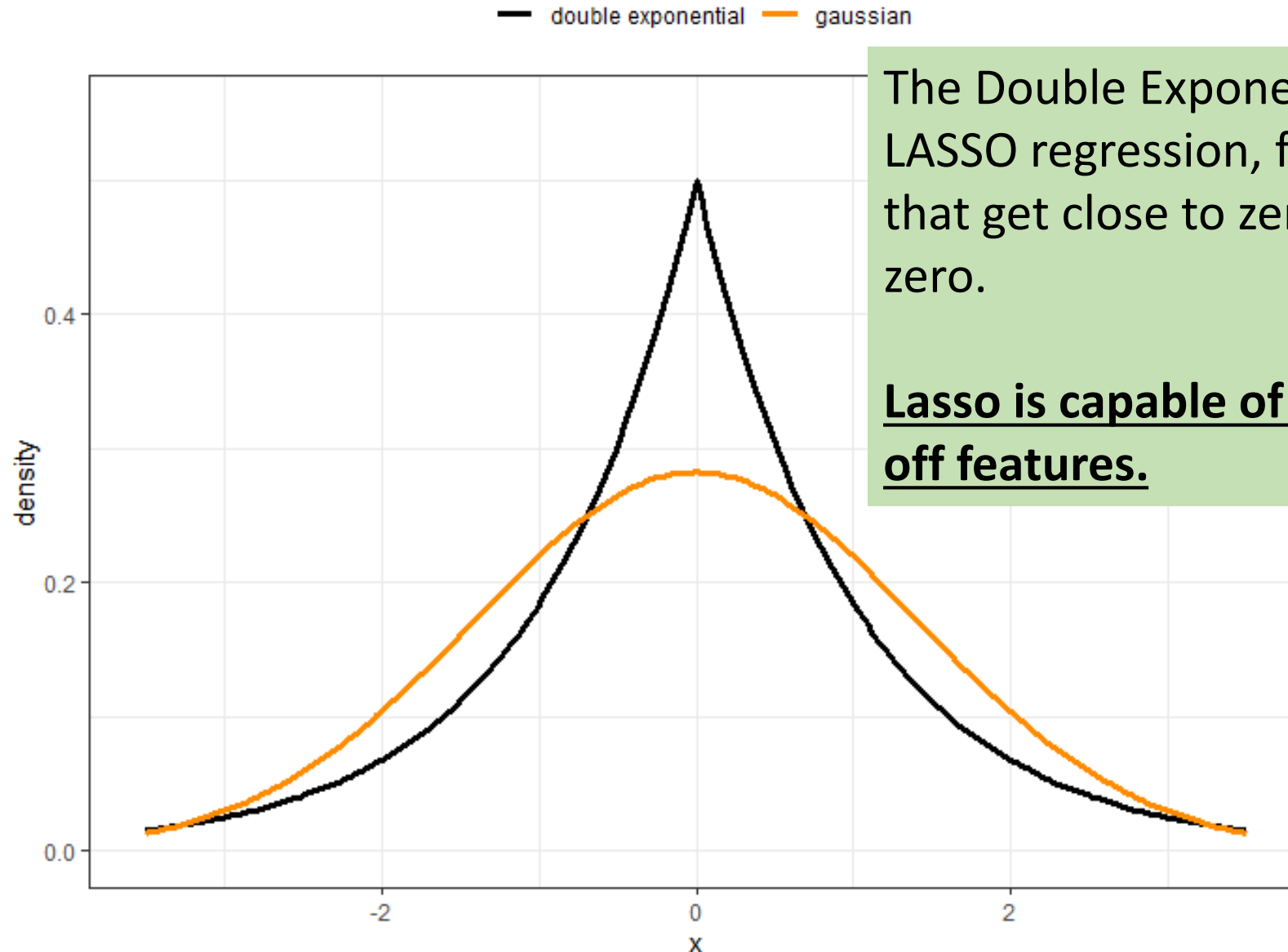
Compare the Double Exponential pdf to a Gaussian with the **same variance**



Compare the Double Exponential pdf to a Gaussian with the **same variance**



Compare the Double Exponential pdf to a Gaussian with the **same variance**



The Double Exponential, and thus LASSO regression, forces coefficients that get close to zero to **LOCK** onto zero.

Lasso is capable of literally turning off features.

In addition to Lasso and Ridge...we can blend the two with ELASTIC NET

$$\operatorname{argmin}_{\boldsymbol{\beta}} \left\{ SSE(\boldsymbol{\beta}) + \lambda \left((1 - \alpha) \sum_{d=0}^D (\beta_d^2) + \alpha \sum_{d=0}^D (|\beta_d|) \right) \right\}$$

The α parameter controls the mixing between the two.

The α parameter must be turned through cross-validation along with λ

Lasso, Ridge, and Elastic Net are implemented in the `glmnet` package in R

- Provides functionality to tune the penalty term, λ , for a given mixing weight, α .
- Easier to tune both λ and α with packages like `caret` or `tidymodels`.